PhD notebook

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1 Administration

⇒ PhD planner

≻ Departemental work hours have to be registred in the DTU system, how?, when?

2 Academics

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2.1 PhD courses

27.5 to 30 ECTS have to be accumulated during the PhD.
Courses propositions:

- 41392 Journal club in fluid mechanics, 5 ETCS. Responsible: Jens Honore Walther , Lyngby Campus, Building 403, Ph. (+45) 4525 4327 , jhwa@dtu.dk. Co-responsable: Knud Erik Meyer , Lyngby Campus, Building 403, Ph. (+45) 4525 4337 , keme@dtu.dk
- 41690 Electron Microscopy and Analysis for Materials Research, 5 ECTS. Responsible: Oleg Mishin , Lyngby Campus, Building 425, Ph. (+45) 4525 2267 , olmi@dtu.dk. Co-responsible: Jakob Birkedal Wagner , Lyngby Campus, Building 307, Ph. (+45) 4525 6471 , jabw@dtu.dk
- 41691 Ph.D. course on application of x-ray diffraction in materials science, 5 ECTS. Responsible: Wolfgang Pantleon , Lyngby Campus, Building 425, Ph. (+45) 4525 2315 , pawo@dtu.dk. Co-responsible Karen Pantleon , Lyngby Campus, Building 425, Ph. (+45) 4525 2207 , kapa@dtu.dk
- 41735 Experimental methods in 3D printing (additive manufacturing), 5 ECTS. Responsible: Venkata Karthik Nadimpalli , Lyngby Campus, Building 427, Ph. (+45) 4525 4753 , vkna@dtu.dk. Co-responsible: David Bue Pedersen , Lyngby Campus, Building 427, Ph. (+45) 4525 4810 , dbpe@dtu.dk. From 3rd to 21st of August 2026. Done
- How to write a scientific paper, 2.5 ECTS. Teacher: Stuart Wright, AIS Sprog. In process
- PhD-Planner: Supervision and facilitation of learning course (1 ECTS). Email: liv@nordicwritingacademy.dk Registered
- Sustainability evaluation and communication (2.5 ETCS). Email: alau@dtu.dk Registered

3 Bibliography

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4 Review paper

- Gas Atomization for Additive Manufacturing: Controlling Particle Size Distribution and Sphericity through Process, Nozzle Design, and Gas Flow Physics
- Gas Atomization for Additive Manufacturing: Controlling Particle Size Distribution and Sphericity through Process and Nozzle Design
- How to control particle sphericity and size distribution in gas atomization for AM
- From Melt Breakup to Powder Morphology: A Physics-Based Review of Gas Atomization for Particle Size Distribution and Sphericity Control

ABSTRACT:

Gas atomization is widely used for the production of metallic powders for additive manufacturing, where particle size distribution (PSD) and sphericity are key determinants of process performance. This review examines the relationships between process parameters, nozzle geometry, and resulting powder characteristics, with particular emphasis on the control of PSD and particle shape. After a brief overview of powder production routes, the influence of melt conditions, gas environment, and operating parameters on atomization behavior is discussed. Special attention is given to nozzle design as a governing factor of gas flow structure and fragmentation mechanisms. Fundamental aspects of gas dynamics, including compressible flow and nozzle exit velocity, are revisited to provide a physical basis for understanding these effects. The review highlights key mechanisms linking process conditions to powder morphology, identifies current limitations in predictive approaches, and outlines perspectives for process optimization in additive manufacturing.

INTRODUCTION:

Since its commercialization in the late 20th century, additive manufacturing (AM) has experienced rapid growth, leading to a significant increase in the demand for high-quality metallic powders [1]. Numerous processes have been developed to produce metallic powders, each offering different levels of productivity, cost, and powder quality, which inherently determine their suitability for specific applications.

Among the key powder characteristics, particle size distribution (PSD) and particle sphericity play a central role, as they directly influence powder flowability, packing density, and process stability. In powder bed-based AM processes, such as laser powder bed fusion, variations in powder morphology can lead to poor layer uniformity, defect formation, and variability in the mechanical properties of the final components. Consequently, controlling powder characteristics during production is of critical importance.

Gas atomization has emerged as the predominant method for producing metallic powders for AM applications due to its flexibility, scalability, and its ability to generate relatively spherical particles with controlled size distributions. Over the past decades, extensive research has been conducted on various atomization processes, including vacuum induction melting gas atomization (VIGA), electrode induction gas atomization (EIGA), plasma-based processes, as well as alternative routes such as water atomization and centrifugal atomization. While these studies provide a broad understanding of available technologies, the relationships between processing conditions and resulting powder characteristics remain only partially understood.

In particular, most existing works address process parameters such as melt temperature, gas pressure, and material properties without systematically linking them to the underlying physical mechanisms governing droplet breakup and solidification. As a result, predictive control of PSD and particle sphericity remains limited, and process optimization is still largely based on empirical approaches.

A critical aspect contributing to this limitation is the insufficient consideration of gas flow behavior and nozzle design. The geometry of the atomization nozzle strongly influences the structure of the high-velocity gas flow, including compressibility effects, turbulence intensity, and the interaction with the molten metal stream. These factors directly control momentum transfer, jet stability, and fragmentation regimes, ultimately determining particle size distribution and morphology. Despite its importance, the role of nozzle geometry has not been comprehensively addressed across different atomization processes.

To address these challenges, this review provides a structured analysis of gas atomization with a focus on the control of PSD and particle sphericity through process parameters and nozzle design. After a concise overview of the main powder production routes, the discussion focuses on the key factors influencing atomization behavior, including melt conditions, thermo-physical properties,

and gas environment.

In order to support this analysis, fundamental aspects of gas behavior are revisited, including ideal gas properties, specific heat ratios, speed of sound, and compressible flow in nozzles. These elements are introduced to clarify how operating conditions determine gas velocity at the nozzle exit and, consequently, the efficiency of momentum transfer and droplet fragmentation. This approach provides a consistent physical framework linking process conditions, nozzle flow characteristics, and powder morphology.

Finally, this review synthesizes current knowledge on the relationships between process parameters, gas flow, and particle formation, highlighting the key mechanisms governing PSD and sphericity. Existing limitations and inconsistencies in the literature are discussed, and perspectives for improved process control and nozzle design are proposed, with particular relevance to additive manufacturing applications.

4.1 Powder production processes

Since its commercialization int the 1680s, the additive manufacturing market growth and consequently the demand for high quality metallic powders has increased significantly [1]. Many processes have been developed to produce metallic powder, with different yield and quality, which naturally dedicate them to different applications.

4.1.1 Vacuum Induction melting Gas Atomization (VIGA)

The gas atomization has been introduced in the 1980s by Crucible Materials Corporation [2]. The VIGA is nowadays the most common process for producing metallic powders for additive manufacturing. The feedstock is melted in a crucible by induction heating, then atomized by high velocity gas jets. The atomization can be done in a vacuum or in an inert gas atmosphere, which allows to produce high quality powders with low oxygen content and good sphericity. The VIGA process is divided in two categories: the free fall and confined gas atomization. The main difference is the distance between the bottom of the liquid metal nozzle and the gas impact point. On the free fall side, the metal stream is hit after a certain distance (d_1) on Figure 1 a), which can be more than 100 mm under the delivery tube [3]. An important distance between the liquid metal nozzle and the interaction point helps to prevent the backflow and freeze-off risk. However, the metal stream properties is less optimal: it becomes larger because of the surface tension, faster due to the gravity, colder because of the heat loss during the free fall, which can lead to a less efficient atomization and a larger powder size distribution. Also the d_2 distance is usually bigger in the free fall configuration, which can lead to a less efficient atomization since the gas jet is less aggressive on the liquid metal stream.

On the confined side, the gas jet is hit the liquid metal stream just after it leaves the delivery tube (generally tangential to the liquid metal stream [4]) which optimizes the maximum metal-gas interaction then ensure small droplet size. However, having the gas nozzle very close to the liquid metal delivery tube increases the risk of backflow and freeze-off.

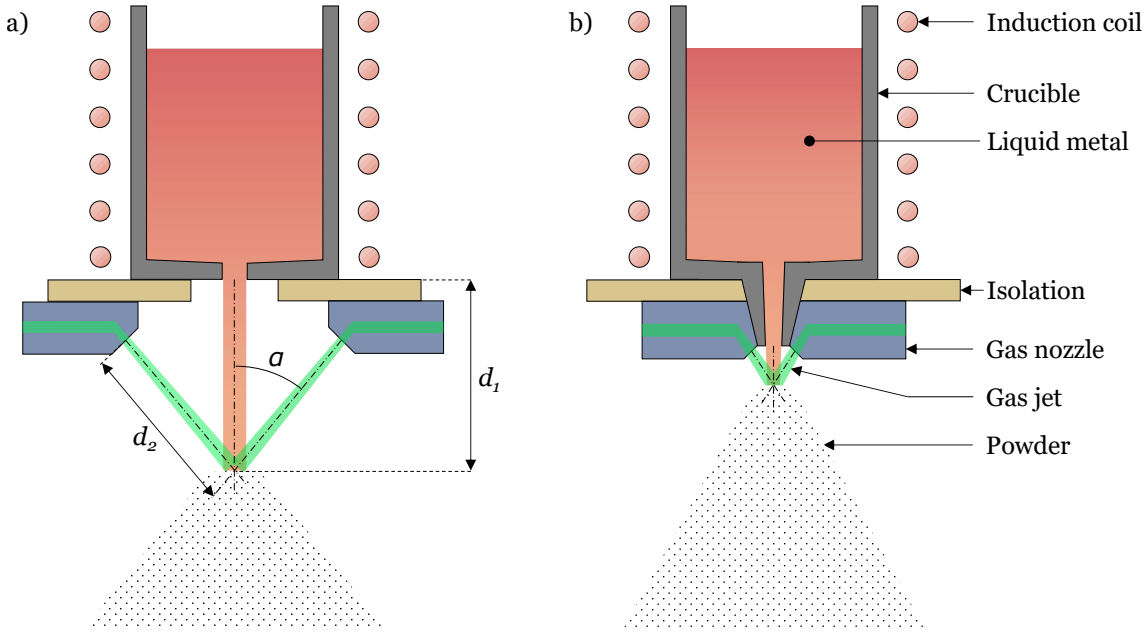

Figure 1: Free fall and confined gas atomization. a) Free fall gas atomization b) Confined gas atomization. α is the apex angle between the liquid metal stream axis and the gas jet axis, d_1 is the vertical distance between the bottom of the crucible and the gas impact point, and d_2 is the distance between the gas nozzle output and the impact point. Adapted from [5].

≻ Why: gas jets direction are generally tangential to the liquid metal stream

If the production cost and rate is quite appreciated in the industry, this process still have some limitations. At first, the powder tends to have satellite particles which has a negative impact on the powder flowability that demanding printing processes such as Laser Powder Bed Fusion (LPBF) requires. Also, hte inert gas used for the atomization may be trapped in the droplet and create porosity in the powder particles. Finally, the powder size distribution is quite wide and the yield of the fine powder is quite low (approximately 15% for the Free-Fall Gas Atomization. The confined gas atomization improves the yield of fine powder and its quality [2, 6]).

The Weber number explains this phenomenon. It is the ratio of the inertial forces of the gas to the surface tension forces of the liquid metal and is given by the formula [7]:

$$We = \frac{\rho_g v_g^2 d_m}{\sigma_m} \tag{1}$$

Where:

ρ_g [kg/m^3]: density of the gas

v_g [m/s]: velocity of the gas

d_m [m]: diameter of the liquid metal stream

σ_m [N/m]: surface tension of the liquid metal

When We < 0, the liquid metal will not break up and fall continuously as a stream. When We > 0, the liquid metal will break up into droplets. The higher the Weber number, the more likely

the liquid metal will break up into smaller droplets [8].

To make the droplet as small as possible, the Weber number has to be as high as possible:

- Increase the gas density ρ_g (Section 4.2.3 proposes a table with different gas and their density, price, reactivity, etc. Possible combinations?)
- Increase the gas velocity v_g (caution to backflow and freeze-off risk)
- Increase the number of gas jets (limited by gas cost, caution to backflow and freeze-off risk)
- Decrease the diameter of the liquid metal stream d_m (limited by the feedstock melting process and the risk of clogging)
- Decrease the surface tension of the liquid metal σ_m (limited by the chemistry of the alloy and the melting process)
- Adjust the angle of the gas jets to be more aggressive (horizontal?) (caution to the backflow and freeze-off risk)

The Reynolds number (Re) is the ratio of the initial force to the gas viscous force. The higher the Reynolds number, the more turbulent the flow. In addition, the Reynolds number is often used in a context of secondary atomization, where the main metal stream is already transformed into coarse multiform particles and thin ligaments. Since this primary atomization result has smaller diameter than the main stream diameter, the Reynolds number becomes even greater in the secondary atomization stage. [9, 6].

$$Re = \frac{\rho_g v_{rel} d_m}{\mu_g} \tag{2}$$

Where:

ρ [kg/m^3]: fluid density

v [m/s]: relative velocity between gas and liquid

d [m]: diameter of the liquid metal stream

μ [Pa.s]: dynamic viscosity of the gas

The Ohnesorge number (Oh) is basically the ratio of the square root of the Weber number to the Reynolds number. It relates the liquid metal stream resistance from being yielded. The higher the Ohnesorge number, the more resistant the liquid metal stream is to break up. By opposition, the gas easily overcomes viscous resistance when the Ohnesorge number is low.

$$Oh = \frac{\sqrt{We}}{Re} = \frac{\mu_l}{\sqrt{\rho_l \sigma_l d_m}} \tag{3}$$

Where:

μ_l [kg·m^{-1}·s^{-1}]: dynamic viscosity of the liquid metal

ρ_l [kg/m^3]: density of the liquid metal

σ_l [N/m]: surface tension of the liquid metal

d_m [m]: diameter of the liquid metal stream

4.1.2 EIGA and PIGA (Electrode Induction / Plasma Inert Gas Atomization)

Both EIGA and PIGA are designed to avoid the contact with ceramic refractive materials such as the crucible. In the EIGA process, the feedstock is a suspended metallic wire which is melted be induction. For a better melting uniformity, the electrode is rotating about 1 to 10 rpm around its axis [10]. The molten metal is then atomized by a high velocity gas jet (see Figure 2 a)). In the PIGA process, the feedstock is commonly a thin wire which is atomized by plasma torches. [3, 11] (see Figure 2 b).

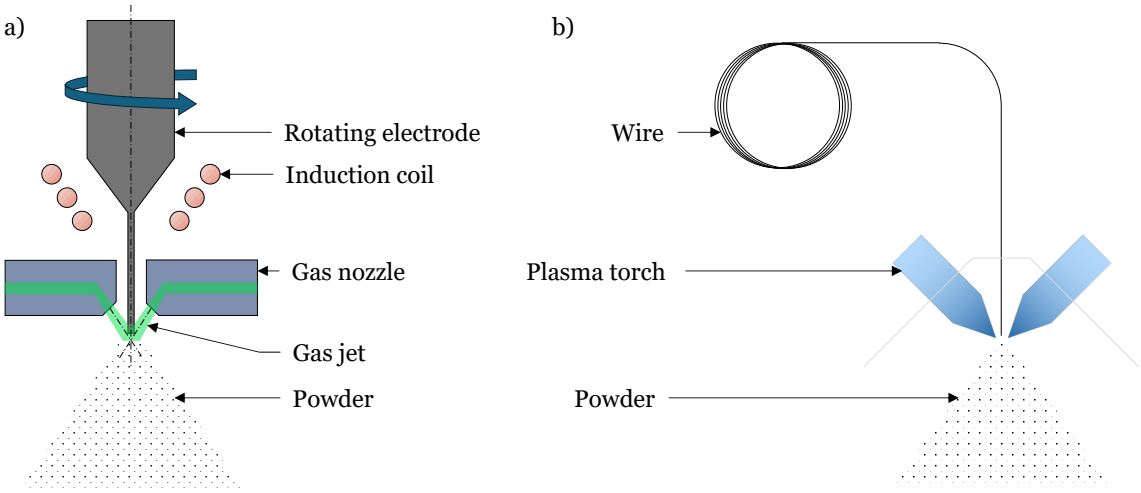

Figure 2: a) EIGA process. Adapted from [10]. b) PIGA process. Adapted from [11, 12].

EIGA is considered as one of the leading processes in the production of Titanium, Zirconium, Niobium and precious metal alloys which can be very reactive with ceramics. The yield of the fine powder and the quality of the produces particles are greater than those obtained by gas atomization. However, the porosity and presence of satellites still need to be improved [2, 13]. Dion et al. propose to add an induction coil around the wire before the plasma atomization, in order to heat it below the melting point. This pre-heating can increase the maximum capacity about 3 or 4 times compared to a system without induction coil [14].

4.1.3 Plasma Rotating Electrode Process (PREP)

In this process, illustrated in Figure 3, the feedstock is a rotating cylindrical electrode which is melted by plasma and atomized by the centrifugal force. The molten metal flies radially and forms spherical particles when it solidifies. The parameters such as the rotating speed, the electrode diameter, the plasma power directly influence the particle size distribution. [15]. As an example, Liu et al [16] have developed a supreme-speed PREP where the electrode rotation can achieve 28,000-36,000 rpm (the standard rotation speed being 3,000-15,000 rpm [2]). As a results, the yield has been increased of the production of fine powder (<45 μm) comparing to the conventional PREP (where the particle size of Ti-6Al-4V is approximately 100-300 μm [2]). This process gives a small porosity since it doesn't involve high pressure inert gas. Also, The probability of particle interaction with each other is very low so the amount of satellites decreases consequently. One of the drawback of this process is the resulting particle size which is not suitable for powder-bed-fusion additive manufacturing. Some improvement are proposed by Zhao et al. [15] gaz nozzle can be added around the rotating electrode to create a gas blast which act on the refinement of the

powder.

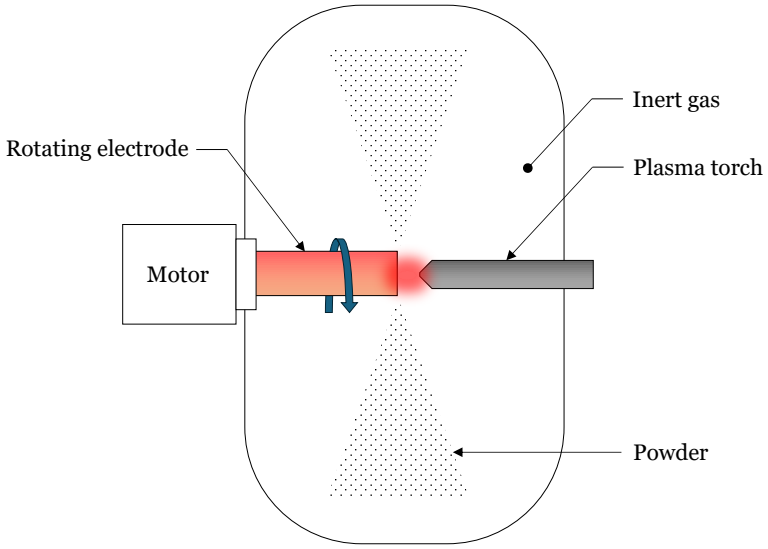

Figure 3: Plasma Rotating Electrode Process (PREP). Adapted from [12, 15, 17].

4.1.4 Water atomization

Water atomization is a process where the molten metal is atomized by high-pressure water jets. The water is usually injected through multiple nozzles, quite similarly to gas atomization. This process is generally used for producing powders dedicated to sintering applications, where the powder is not required to be spherical and can have a wide size distribution and injection moulding. The cheap price of water and the high cooling rate of the droplets make this method very competitive for producing large quantities of powder. However, the powder quality is lower than gas atomization, with a higher oxygen content and a more irregular shape. The water atomization installation is composed of an induction furnace, an atomization chamber, a dehydration stage, a drying stage and a powder collector.

Y Seki and al [18] studied two different nozzles for water atomization, the cone and the V-jet, illustrated in Figure 4. The cone nozzle is the most common in gas atomization, producing a conical spray focused on the metal stream. The V-jet nozzle produces two flat sprays joined in a V shape, which creates either a better suction and a better slicing of the metal stream. The reason why the suction effect is better is that the V-shape let more space for the pressure to escape while the conical one contains it in the cone, allowing a less important depression. However, a cone nozzle is much better for obtaining spherical particles, since it creates a more uniform and stable atomization environment. This is the reason why the V-jet are not used for gas atomization, an axi-symmetrical nuzzle is preferred.

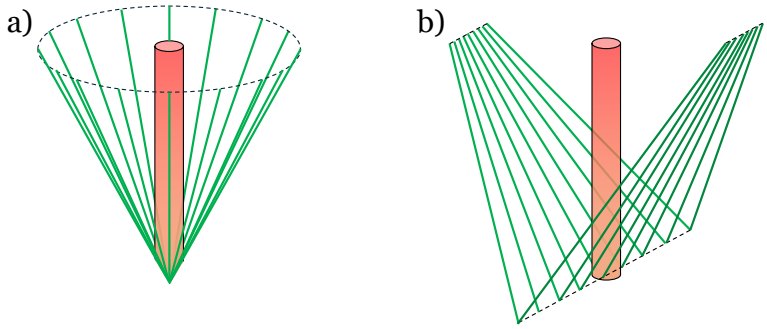

Figure 4: Cone and V-jet nozzles for water atomization. a) Cone nozzle b) V-jet nozzle. Adapted from [18, 19].

Figure 4 shows that the V-jet nozzle creates a much higher suction pressure effect than the cone nozzle, which is a clear beneficial factor to decrease the mean particle diameter [18].

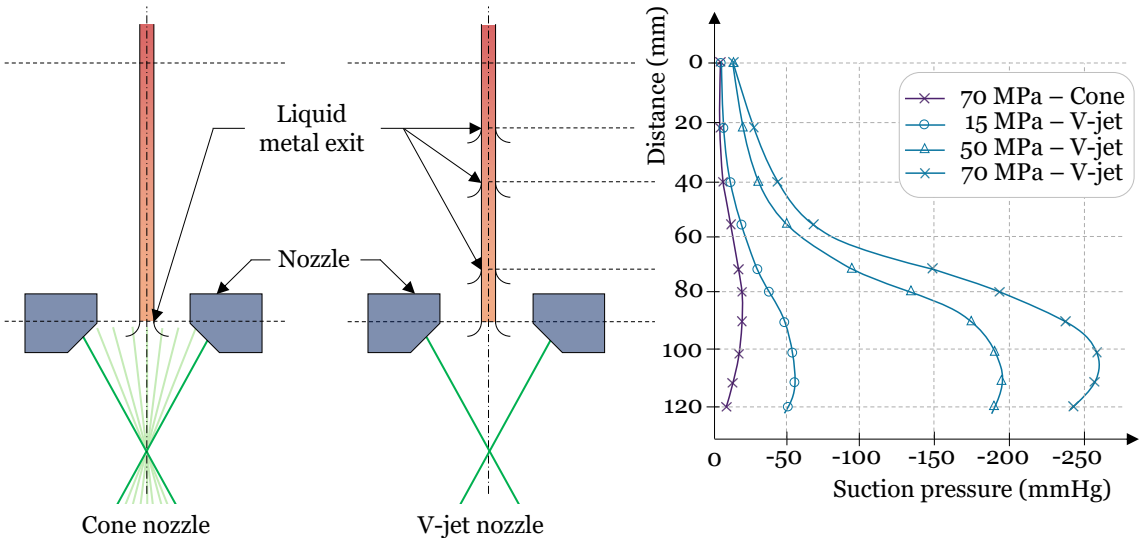

Figure 5: Influence of suction pressure with different nozzles for water atomization. The values -100 and -200 mmHg corresponds respectively to -0.133 and -0.267 bar. Illustration adapted from [18].

The difference in the suction pressure between the two nozzles is significant. With 70 MPa water pressure, the liquid metal exit located at 100 mm from the "zero position", the suction pressure is about -0.033 bar with the cone nozzle and -0.36 bar with the V-jet nozzle.

4.1.5 Centrifugal atomization

The centrifugal atomization has been developed from the spray forming process and is an optimized process to produce powder with a spherical morphology and a narrow particle size distribution. Since only the centrifugal force is used to atomize the jet, there is no porosity due to gas-entrapment and the costs is reduced. The molten metal fall on a spinning disc or cup which has a rotation speed of about 20,000-30,000 rpm [17, 20, 21]. Some studies have shown that the yield increases with the rotation speed (Suastiyanti et al. [21] for Sn and Li et al. [22] for aluminum).

However, Wu et al. [17] raise a transition from a unimodal to a bimodal particle size distribution when the rotation speed exceed 33,000 rpm. The centrifugal forces stretches the droplets and links them to each other with a necked-down liquid metal ligament. The rotating disc and cup are made of stainless-steel matrix with composite coating. Indeed, the material has to be resistant to high temperature and corrosion, with a good wettability as well as a low thermal conductivity. This way, there is a good contact between the molten metal and the disc but the heat is not quickly extracted from the metal and transferred to the motor.
If the chemical and physical properties of the powder are quite good, the particle size is still too big for the most demanding additive manufacturing processes such as LPBF.

4.1.6 Mechanical attrition

The word "attrition" is used to describe the process of a size reduction when powdered materials are subjected to mechanical forces. The attrition process is commonly used for obtaining powder that are finer than those obtained by atomization processes. The attrition gathers different process as tumbler ball mills, vibratory ball mills, attrition mills, hammer and rod mills, etc [23].
Shinohara et al. [24] have investigated the grinding caracterestics of hard materials as synthetic diamond and alumina. They have proved that the grinding rate of hard material can be increased by optimizing parameters as the size of the milling media, the rotation speed, the filling ratio and the dispersing liquid volume. The equipment is composed of a stainless steel chamber with rotating stirrer system in the middle, which can have a rotation speed between 90 and 528 rpm. Some cylindrical pins (6 mm diameter and 65 mm length) are arranged a 90◦ to each other. The grinding media is steel balls (2.00, 4.76 and 6.36 mm diameter) that fill 80% of the chamber volume. The diamond powder has a median size that decreases as a function of the grinding time, reach 1.5μm after 1h and less than 1μm after 2h.
Fox [25] describes similar high-speed milling process used for a particular purpose: demonstrating that metal hydrides can be produced by mechanical alloying to support the tritium production facilities at the Savannah River Site, operated for the U.S. Department of Energy. It has been demonstrated that elemental metal powder could be alloyed in an attritor mill under argon. This attritor is basically a high speed mill for rapid particle size reduction, composed of a stationary vessel filled with milling media (typically spherical 3-4 mm diameter balls) and the material that has to be ground with a motor driven shaft with horizontal arms. One of the test was a planetary mill with 3/16" diameter tungsten carbide grinding media, operated at 400 rpm during 15-40 hours.
Kilinç et al. [23] have used a vibratory horizontal attritor in order to characterize alumina powders production. The process uses 6.35 mm diameter media stainless-steel balls in an inner vessel, water-cooled thanks to an outer vessel. The system is sitting on four pressure springs. The rotational speed of the attritor has been set at 500, 700 and 900 rpm and an optimum at 700 rpm has been found. The obtained particle size is about few microns (about 2μm after 5h). However, the powders shows a platelike structure with a short milling time and they become nodular and agglomerated with longer milling time.

4.1.7 Ultrasonic atomization

Ultrasonic atomization was born to bring an alternative to produce fine and spherical particles for additive manufacturing. The metal can be molten with plasma torch for high-temperature alloy, induction coil for low-melting alloy [26], or electric arc. Bałasz Et al. [27] investigated the ultrasonic atomization process of 316L stainless-steel. The material is melted by electric arc and positioned on a sonotrode which transmit the ultrasonic vibrations from the generator to the molten metal. Due to these vibrations, the molten metal is ejected from the surface and transported by a high-pressure flow of argon. Priyadarshi Et al. [28] have studied the ultrasonic atomization of aluminum alloy, water, and glycerol. To melt the aluminum, they used an induction coil around a graphite crucible with a 0.5 mm diameter orifice nozzle. Bellow, a CF flange is vibrating at 60 kHz

to atomize the melt, reaching a yield greater than 80 %. As a results, they found that cavitation plays an important role in the atomization process. It helps to create small waves on the liquid surface that break into droplets. The collapse of cavitation bubbles also generates shock waves that further assist in breaking the liquid apart. In addition, cavitation produces a swirling, bubbly flow that helps move and disperse the liquid before it atomizes. The study also showed that process parameters, such as vibration amplitude, liquid thickness, and wetting, affect the size and shape of the produced particles. Ciftcli et al. [26] have used ultrasonic atomization to produce powder with niobium C103 (Nb-10Hf-1Ti) alloy, magnesium alloy modified with silver and recycling of the titanium machining chips, on rePOWDER from AMAZEMET, Poland. The magnesium alloy was melted by induction coil and atomized on a blade vibrating at 60 kHz. The obtained powder has a spherical morphology with a good size distribution but with some sub-micron particles coming from possible zinc evaporation and particle bonding. Indeed, the melted temperature has to be set at 1223 K to ensure a stable flow, however the zinc boiling point is 1180 K. The Ti-alloy chips from CNC machine have been melted using plasma torch into ingots and atomized at 40 kHz. They have obtained a very small amount of large particles, with over 85 % of the powder is below 100μm.
Li Et al. [29] have used this technology to produce high-thermal-conductivity, low-melting-point In–15Pb–5Ag solder paste, achieving fine spherical powder, low porosity, and improved heat dissipation for advanced electronic packaging. Using a graphite crucible, an induction coil and a vibrating ultrasonic horn, they have tested different configuration: keeping a melting temperature of 300°C and an ultrasonic frequency of 40 kHz, they tried different vibration amplitudes (40, 60 and 80 %). 40 % was the optimum for the powder sphericity. Above, more energy is transferred to the melt forming messy metal droplets. Wisutmethangoon Et al. [30] have also used ultrasonic atomization to produce SAC305 with a very similar process. They found that increasing the melt temperature, while decreasing the melt feed rate and vibrational amplitude, reduces both the median particle size and the particle size distribution. Additionally, lowering the oxygen content in the atomizing chamber leads to more spherical SAC305 powder and reduces the oxygen content on the powder surface.

4.1.8 Physical Vapor Deposition (PVD) and Chemical Vapor Deposition (CVD)

The PVD is part of the gas-phase method which allows to produce very fine and high-purity powders. During the first stage, the metal is melted until being converted into vapor. The high temperature can be achieved using various heating methods: resistance or induction furnaces, electron beams, low-voltage arcs, hollow cathodes, or laser beam heating. The vapor is then transported towards a condensation zone. The condensation can happen either in a free space or on a cooled surface to transform the vapor in to powder. This process requires controlled atmosphere with a neutral gas, with a presence of oxidant or passivating agent. In the reference [19], a installation for zinc powder is described, where a 100 kW inductor is needed at each section of the system, with a up to 300 kg capacity of molten zinc in a graphite crucible. The zinc vapor is transported in the nitrogen flow toward the condensation zone, in a rotating water-cooled drum. The powder is collected with a blade and dropped in a bunker. The units yield is about 10-15kg/h of ultra fine zinc powder. Another machine called "Tuman" is also described, developed by the Institute of Metallurgy, Ural Branch of the Russian Academy of Science. The annual powder production rate is about 120 tons/year of zinc powder. The produces particles of these process can reach the nano-scale. The nano-powders are usually used for specific fields such as production of rocket fuel, catalysts, heat absorbents and microelectronics. However, the process is complex and expensive with a low-to-moderate production rate, which limits its industrial applications.

CVD is less common for powder production than coatings application. However, powder par-

ticles are obtained with the reaction of gaseous precursors in the vapor phase. The by-products of the reaction if removed by gas flow. The powder formation in controlled by the type of precursors, the degree of supersaturation, the temperature and the composition of the gas atmosphere. As PVD, the this method allows very fine (nano-particles) [19].

4.1.9 Cross-disciplinary and combined processes

4.1.9.1 Rocket fuel atomization

Another field of application for atomization with very optimized processes is the injectors of rocket engines. The atomization of the fuel is a critical step for the combustion efficiency and the performance of the engine. The smaller the droplets, the bigger the surface area for the combustion, which leads to a more efficient combustion. Casiano Et al. [31] describe different Liquid-Propellant Rocket Engines throttling technics. Among the different devices, a recurrent process is the use of a fuel vessel ending to a multiple of nuzzles, which is imbricated in a oxygen vessel itself ending to a multiple of nuzzles, interacting with the fuel onces. A very common way is to position these nozzle as "triplet", where the fuel nozzle is perpendicular to its supporting flange, and two oxidizer nozzles are located on both sides, on the same line, converging towards a single interaction point with the fuel flow. This can be compared to a multi-outlet conical gaz nozzle cross section of a metal atomizer, where the metal stream is the fuel flow, and only two gas jets are visible on the section. Kang Et al. [32] present a review on pressure swirl injector for rocket engine. On that case the atomization is done by the centrifugal force created by the swirling motion of the fluid. The fuel is injected tangentially into a converging spin swirl chamber, creating a vortex that causes the liquid to spread out and form a thin sheet (primary breakup). The swirling film becomes unstable and breaks up into droplets (secondary breakup). Yu ET al. [33], Copenhagen Suborbitals [34], Radke ET al. [35] and Sivakumar ET al. [36] describe a liquid-liquid double swirl coaxial injector for rocket engine applications, in which the fuel and liquid oxygen are injected through concentric nozzles and imparted with angular momentum prior to injection. This generates concentric, rotating liquid sheets upon exit, which expand into thin conical films that interact dynamically, leading to rapid breakup into fine droplets and enhanced mixing between the propellants. By designing the length differential between both the inner and outer parts, the interaction area can be controlled. Compared to more complex rocket injector designs, these coaxial swirl nozzles are relatively simple to manufacture while still providing effective atomization performance. Figure 6 illustrates a single double swirl injector, knowing that one rocket engine has multiple injectors.

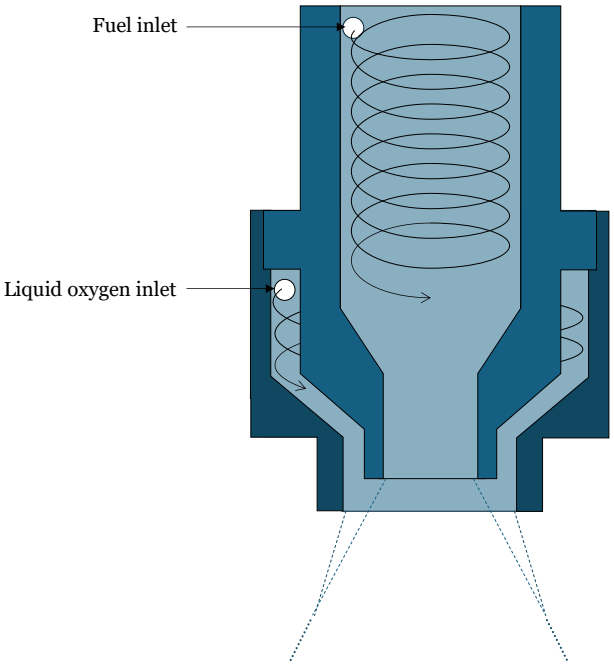

Figure 6: Liquid-liquid coaxial swirl injector

4.1.9.2 Spray forming

Spray forming is a process that employs gas atomization to produce metallic powder. The powder is subsequently deposited onto a rotating disc just before it solidifies. This disc is also translated to maintain a constant flying distance between the melt nozzle and the growing billet surface. Owing to the use of a fine powder feedstock, the resulting billet exhibits improved properties compared with conventionally cast material. Indeed, all powder particles experience similar conditions upon arrival at the billet, combined with a very high cooling rate, which reduces chemical segregation and promotes a more homogeneous microstructure. This technique is widely applied to stainless steels, superalloys, cast irons, tool steels, as well as aluminum and copper-based alloys [37, 38].

4.1.9.3 Combined processes for metal powder production

Neikov Et al. [19] describe different processes which gather several atomization methods. As an example, the "pressure-swirl hybrid pre-filming atomizer" is based on a swirling molten metal injected tangentially in a melt nozzle, leaving this nozzle by forming a conical sheet. The pre-breakup can be compared to the phenomena present in the inner part of the coaxial swirl injector (see Figure 6). Then, the conical sheet is hit by high pressure gas jet. Another example is the "pre-filming hybrid atomizer", dedicated to high viscosity melts. In this process, the molten metal is dropped on a rotating disc (centrifugal atomization) and then atomized with a high pressure gas jet.
Qiu et al [39] propose a atomization process combining the centrifugal, the oscillating and the impact breakage atomization. The idea is to flow the molten metal on a rotating disk which is also oscillating, and break the centrifuged droplet by hitting them against fine threads attached on another disk which is rotating in the opposite direction.

4.1.10 Powder quality comparison of production routes

This chapter has provided a comprehensive review of the main fabrication routes, highlighting their respective principles, advantages, and limitations. The comparison of these approaches emphasizes the strong dependence of the final material properties on the processing conditions and underlying mechanisms. To facilitate a global understanding of these relationships, a summary plot is introduced in Figure 7, synthesizing the key features discussed throughout this review.

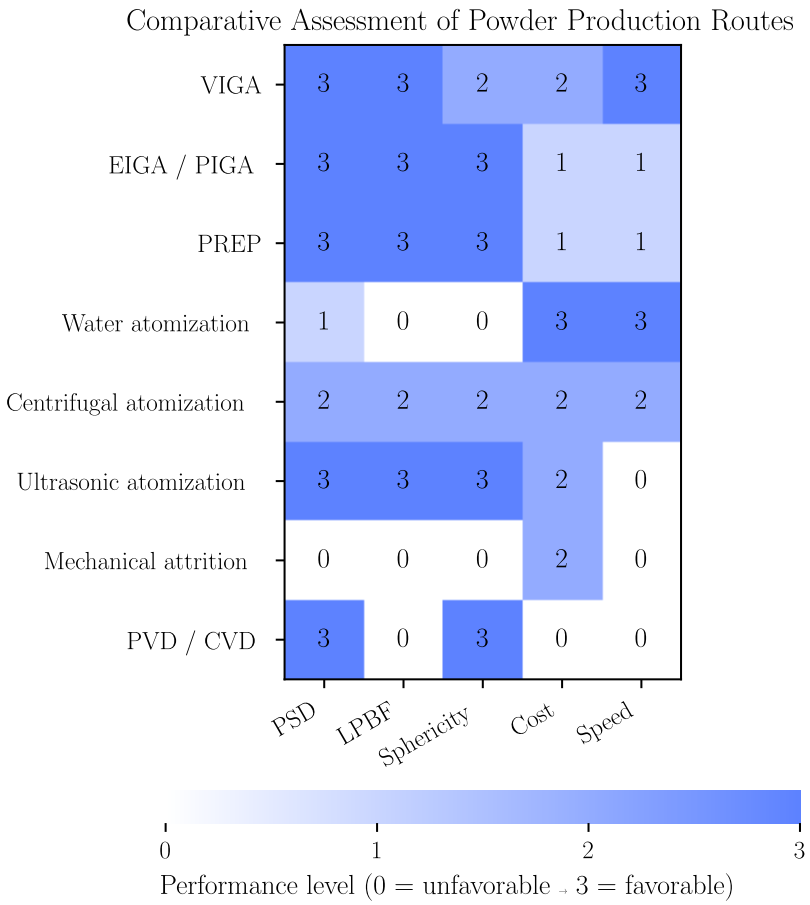

Figure 7: Comparative Assessment of Powder Production Routes

4.2 Factors influencing metallic powder size and quality during atomization

4.2.1 Feedstock melting

One of the melting processes is to use a crucible heated by induction. It allows different types of feedstock such as powder, scrap, wire, and can accept pure metal elements or pre-alloyed elements. All these parameters will impact the powder quality. For instance, the melted scraps can be heterogeneous and may contain oxides and impurities. In these cases, it is recommended to take a sample of the homogeneous melted material in order to analyze the chemistry [3].

Both open and closed melting systems are used in gas atomization. When the metal is molten in open air, the risk of oxidation and contamination is increased, even if the slag provides a natural

protection that is very commonly used in pyro-metallurgical processes [40]. Open air induction heating system can accept a bigger volume of metal but it will give a less good powder quality. In other processes, the metal is molten with plasma torches or electric arc.

In a context of process optimization, characterizing different feedstocks with the on-site facilities would be a nice sanity check to align with the literature results. Eventually, once this first step is controlled, it would be nice to continue the parameters optimization with vacuum heating. In order to be able to analyze properly the results, only one parameter has to be changed for each iteration. To decrease the results given by combined parameters and to be able to improve the powder quality, on way would be, when it is possible, to focus on one parameter, optimize it, and keeping it optimized while modifying the next parameter, one by one. Some of the parameters are known in the literature to be significant in the powder quality improvement as well as requiring complex or expensive investments such as helium environment or a high quality feedstock. These parameters should be punctually tested for the sanity test and kept in mind to be re-introduced in the process only when it is consistent.

4.2.2 Thermal condition and properties of the molten metal

The temperature of the molten metal and its surface tension affect the process efficiency. When increasing both the gas and liquid metal flow, the powder size distribution is reduced [3].

≻ PhD Daniel, Figure 2.9: Effect of temperature gradient G and growth rate R on the morphology and size of solidification microstructure. Reproduced from [83].

4.2.3 Gas environment

In the atomization process, the gas environment is a critical parameter for controlling powder quality. The choice of gas is generally driven by cost considerations. Furthermore, the gas also serves multiple important functions in the process:

- create a protective atmosphere to prevent oxidation and contamination
- control the feeding of the molten metal
- atomize the molten metal
- control the cooling rate
- transport the powder

The often protocol is to flush the chamber(s) of the atomizer with an inert gas in order to remove the oxygen and moisture. The environment can be filled again with an inert gas and maintained just above the atmospheric pressure to prevent the air from entering in the system. By adjusting the pressure in the different chambers, the feeding of the molten metal can be controlled As an example, Priyadarshi Et al. [28] filled the furnace and the atomization chamber with argon to 100 mbar before the pouring. The molten metal flow (aluminum in that case) was regulated by maintaining the appropriate differential pressure (initiated at 200 mbar). A turbo pressure of 1.5 bar has been imposed when the nozzle started clogging.

Table 1 proposes a list a inert gases with some properties that can be relevant for the atomization process.

Property	**Helium**	**Neon**	**Argon**	**Krypton**	**Xenon**	**Nitrogen**
Molar mass M (g/mol)	4.00[a)]	20.18[a)]	39.95[a)]	83.8[a)]	131.3[a)]	28.0[a)]
Critical temp. T_{cr} (K)	5.2[a)]	44.5[a)]	150.7[a)]	209.5[a)]	289.7[a)]	126.2[a)]
Critical pressure P_{cr} (MPa)	0.2275[a)]	2.678[a)]	4.863[a)]	5.51[a)]	5.84[a)]	3.396[a)]
Critical mol. vol. V_{cr} (cm^3/mol)	57.48[a)]	41.87[a)]	74.59[a)]	92.25[a)]	118.28[a)]	89.41[a)]
Critical density ρ_{cr} (kg/m^3)	69.64[a)]	481.9[a)]	535.6[a)]	908.4[a)]	1110[a)]	313.3[a)]
Specific heat ratio [γ]						
0.1 MPa, 283 K	NA	NA	NA	NA	NA	1.4[d)]
0.1 MPa, < 400 K	1.67[b)]	1.67[b)]	1.67[b)]	1.67[a)]	1.67[a)]	NA
2 MPa, 400 K	1.67[c)]	1.67[c)]	1.7[b)]	1.72[b)]	1.79[b)]	NA
2 MPa, 1200 K	1.67[c)]	1.67[c)]	1.67[c)]	1.7[c)]	1.79[c)]	NA
7 MPa, 400 K	1.66[c)]	1.67[c)]	1.77[b)]	1.88[b)]	2.21[b)]	NA
7 MPa, 1200 K	1.66[c)]	1.66[c)]	1.66[c)]	1.67[c)]	1.69[c)]	NA
Reactivity	1	2	3	4	5	6
Price	3	4	2	5	6	1

Table 1: Summary table of gas properties. a) [41], b) [42], c) Data visualization reading from [42], d) [43]. For the reactivity and price, a 1 to 6 scale is proposed as a general information, "1" means lowest price and reactivity, "6" means highest price and reactivity. The accurate values depend on other parameters such as the purity of the gas. The nitrogen is chemically stable but can be reactive at high temperature.

As shown in Table 1, the specific heat ratio decreases with the temperature and increases with the pressure. At atmospheric pressure and T<400 K, the specific heat ratio of the noble gases is around 1.67, which is the theoretical value for monatomic ideal gases, with a maximum deviation of 0.3% for xenon ($\gamma = 1.672$). For 2 MPa and 400 K, larger deviation occurs for Argon, Krypton and Xenon with $\gamma = 1.697$, 1.724 and 1.786 respectively, which corresponds to 1.8%, 3.4% and 7.1% from the theoretical value. For 7 MPa and 400 K, the specific heat ratio raises respectively to 6.2%, 12.5% and 32.8% with $\gamma = 1.770$, 1.875 and 2.214. At 7 MPa and 1200 K, the specific heat ratio of the noble gases is comes back close to the theoretical value, with a maximum deviation of 1.8% for Xenon [42].

In a context of simulation and experiment, the theoretical values can be used as a first approach, with a uncertainty could be around 30% for Xenon at high pressure and "low" temperature.

The price order of the gases are in the order of atmosphere respective ratio as shown in Table 2, except of the helium, which is usually extracted in natural gas reservoirs where it can represents up to 7% of the mixture (much more concentrated than in the atmosphere).

Component	Percentage by volume	Percentage by mass
Nitrogen, N_2	78.08	75.53
Oxygen, O_2	20.95	23.14
Argon, Ar	0.93	1.28
Carbon dioxide, CO_2	0.031	0.047
Hydrogen, H_2	5.0×10^{-3}	2.0×10^{-4}
Neon, Ne	1.8×10^{-3}	1.3×10^{-3}
Helium, He	5.2×10^{-4}	7.2×10^{-5}
Methane, CH_4	2.0×10^{-4}	1.1×10^{-4}
Krypton, Kr	1.1×10^{-4}	3.2×10^{-4}
Nitric oxide, NO	5.0×10^{-5}	1.7×10^{-6}
Xenon, Xe	8.7×10^{-6}	1.2×10^{-5}
Ozone, O_3 (summer)	7.0×10^{-6}	1.2×10^{-5}
Ozone, O_3 (winter)	2.0×10^{-6}	3.3×10^{-6}

Table 2: The composition of dry air at sea level [44]

The density of the gas decreases with the temperature and increase with the pressure. The density difference between the different gases is more significant at low temperature.

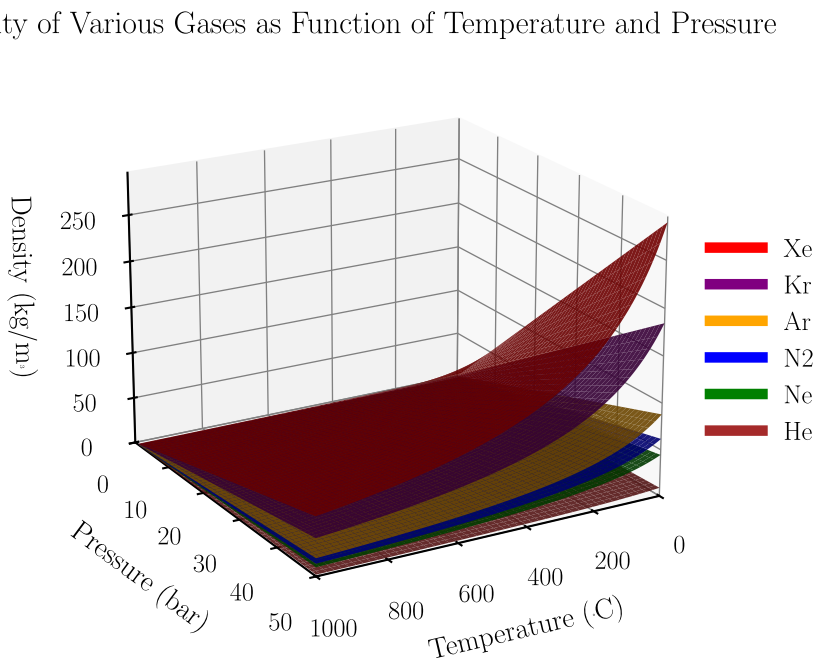

Figure 8: 3D plot of the density of the helium, neon, argon, krypton, xenon and nitrogen as a function of temperature and pressure. Molar mass from [41].

$$c = \sqrt{\frac{\gamma RT}{M}} \qquad with \qquad \gamma = \frac{C_p}{C_v} \tag{4}$$

Where:

c [m/s]: speed of sound

γ: specific heat ratio

R [J/(mol·K)]: specific gas constant

T [K]: absolute temperature

M [kg/mol]: molar mass

C_p [J/(kg·K)]: specific heat at constant pressure

C_v [J/(kg·K)]: specific heat at constant volume

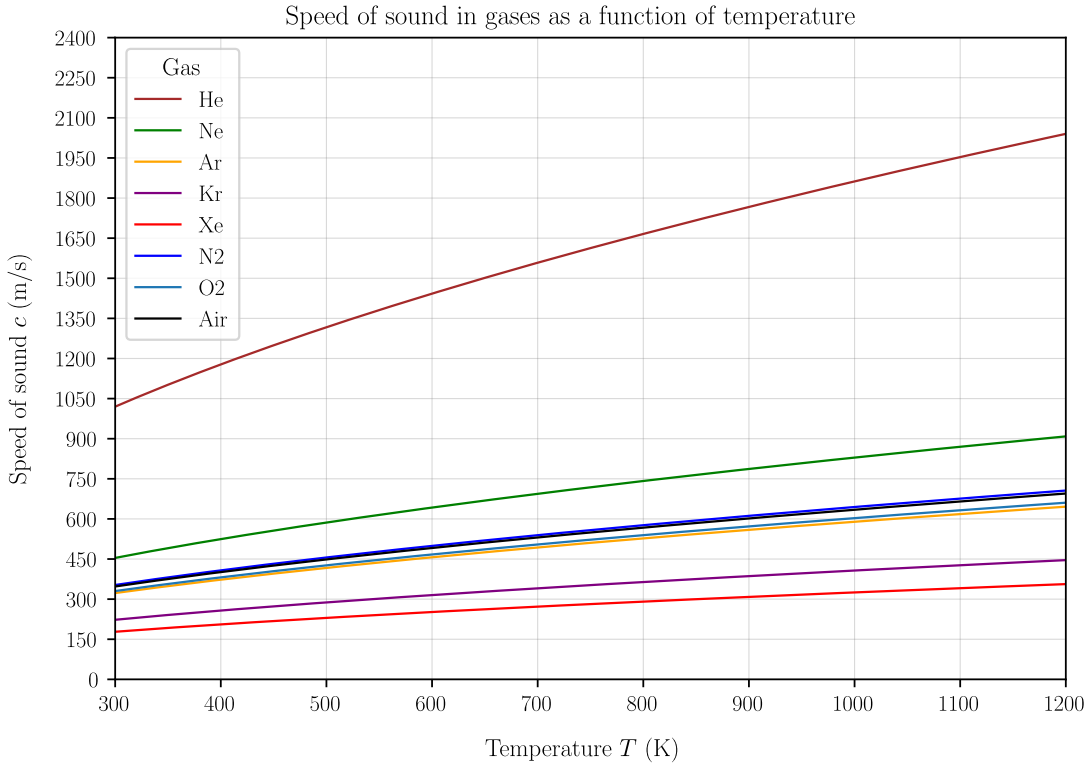

Figure 9: .
Speed of sound in various gases as a function of temperature. The speed of sound is calculated using the Formula 4. The molar mass and specific heat ratio values are taken from Table 1, with γ at 0.1 MPa and >400 K for He, Ne, Ar, Kr and Xe, and γ at 0.1 MPa and 283 K for N_2. The air properties are calculated taking only nitrogen, oxygen and argon, with the percentage of Table 2. M_{oxygen} is given by NIST [45] and γ_{oxygen} is estimated at 1.4 as a diatomic molecule. In this plot, the specific ratio are considered constant as a function of the temperature.

The graph in the Figure 9 shows that the speed of sound is higher in lower molar mass gases and increases with the temperature. When comparing with the speed of sound in air and water, it's confusing that the sound goes faster in a low density gas than is a high density gas, He and N2 for instance. This phenomenon can be explained with several reasons. These two gases have a very different molar mass, the molecules of helium are much lighter than the molecules of nitrogen, which allows them to move faster and transmit sound waves more quickly. Also, their specific heat ratio differ, $\gamma_{He} > \gamma_{N_2}$. This means that He stiffness is greater than nitrogen stiffness, which makes He more reactive to transmite the sound vibration (local pressure). Having a high γ means in that case having a low C_v. Indeed, in a constant volume, a gas cannot expand, so all the energy goes into increasing the temperature. However, in a constant pressure, the gas can expand, so part of the energy is dissipated in the expansion of and the rest goes into increasing the temperature. This is why C_p is always greater than C_v, and consequently why γ is always greater than 1, and this is where C_v - C_p = R. To understand the variation of C_v in the different gases, a focus of the molecular structure is required in order to study their degrees of freedom. The degrees of freedom of a gas molecules are related to their number of atoms and enumerate the number of different way to dissipate energy (translation, rotation, vibration). The Figures 10 shows the degrees of freedom of helium and nitrogen molecules and explains why monatomic gases have fewer degrees of freedom compared to polyatomic gases.

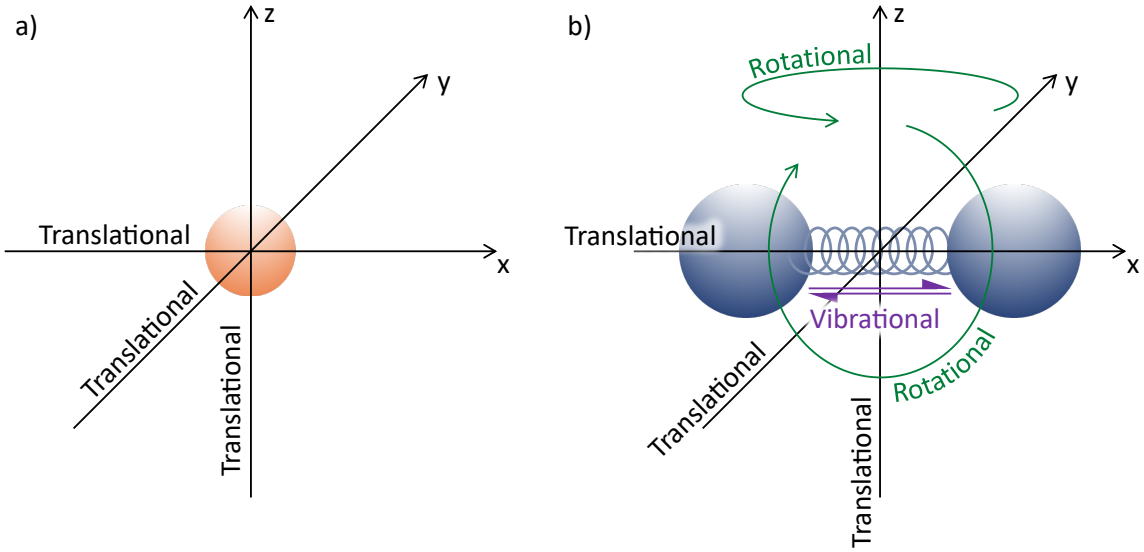

Figure 10: Degrees of freedom of helium and nitrogen molecules. a) Helium is a monatomic gas, it has 3 degrees of freedom (translation). b) Nitrogen is a diatomic gas, it has 5 degrees of freedom (3 translation and 2 rotation). The rotation around the molecular axis is considered negligible since the moment of inertia is almost zero. The vibration degree of freedom is only activated at high temperatures. Adapted from [46].

The motions of a gas molecule are categorized into three types of degrees of freedom: translational, rotational, and vibrational and represent a sum of kinetic and potential energies E_T given by:

$$E_T = E_{translational} + E_{rotational} + E_{vibrational} \tag{5}$$

The translational motion of any molecule is given by the kinetic energy of its center of mass, which is the sum of the kinetic energies in the x, y, and z directions:

$$T_{trans} = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 \tag{6}$$

Where:

m [kg]: mass of the molecule

v_x, v_y, v_z [m/s]: velocity components in x, y, z directions

The rotational motion of a molecule is given by the kinetic energy of its rotation around its center of mass, which is the sum of the kinetic energies of its rotation around the x, y, and z axes:

$$E_{rot} = \frac{1}{2}I\omega_1^2 + \frac{1}{2}I\omega_2^2 + \frac{1}{2}I\omega_3^2 \tag{7}$$

Where:

I [kg·m^2]: moment of inertia

ω_1, ω_2, ω_3 [rad/s]: angular velocity components

For a diatomic molecule such as nitrogen,

$$I = \mu r^2 \tag{8}$$

Where:

r [m]: distance between atomic centers

μ [kg]: reduced mass

The reduced mass is given by the formula:

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \tag{9}$$

Where:

m_1, m_2 [kg]: masses of the individual atoms

In the case of nitrogen, $m_1 = m_2$:

$$\mu_{nitrogen} = \frac{m_{nitrogen}}{2} \tag{10}$$

The vibrational motion of a molecule is given by the sum of the kinetic and potential energies of its vibration around its equilibrium position:

$$E_{vib} = \frac{1}{2}\mu\left(\frac{dy}{dt}\right)^2 + \frac{1}{2}ky^2 \tag{11}$$

Where:

y [m]: vibrational coordinate (displacement from equilibrium)

k [N/m = kg·s^{-2}]: force constant of the bond

To summarize, numerous degrees of freedom induce important capacity to store energy, which gives a high C_v (for a given volume, a lot of energy is needed to raise the temperature since it is partially dissipated in the molecular movements). A high C_v means a low γ and then a low speed of sound.

The speed of the gas in the nozzle is a critical parameter for the atomization process, and depends of the specific heat ratio, the specific gas constant, the temperature and the pressure differential between the gas supply and the atomization chamber. The gas speed can be increased by increasing the pressure differential, which can be achieved by increasing the supply pressure or decreasing the chamber pressure. However, there are practical limits to how much the pressure can be increased due to safety concerns and equipment limitations. The pressure inside the atomization chamber will be just above the atmospheric pressure to ensure that the inert gas is properly contained.

To reach such a formula, the starting point is the energy conservation (Equation 12) and the relation between enthalpy and temperature (Equation 13), from [47].

$$h_0 = h + \frac{v^2}{2} \tag{12}$$

Where:

h_0 [J/kg]: total enthalpy (stagnation enthalpy)

h [J/kg]: static enthalpy

v [m/s]: velocity of the gas

$$\begin{aligned} h &= c_p \cdot T \\ h_0 &= c_p \cdot T_0 \end{aligned} \tag{13}$$

Where:

h [J/kg]: enthalpy

c_p [J/(kg · K)]: specific heat at constant pressure

T [K]: temperature

From these two equations, the following formula can be derived:

$$C_p \cdot T_0 = C_p \cdot T + \frac{v^2}{2} \Leftrightarrow \frac{v^2}{2} = C_p \cdot (T_0 - T)$$

As a results, here is the energy-temperature relation for compressible flow:

$$v^2 = 2 \cdot C_p \cdot (T_0 - T) \tag{14}$$

For the second step, the equation of state (Equation 15) and the isentropic relation (Equation 16) for perfect gases are needed:

$$P = \rho \cdot R \cdot T \Leftrightarrow \rho = \frac{P}{R \cdot T} \tag{15}$$

$$P \cdot \rho^{\gamma} = constant \tag{16}$$

Where:

P [Pa]: Pressure

ρ [kg/m^3]: Density

R [J/(kg $\cdot$ K)]: Specific gas constant

T [K]: Temperature

γ: Specific heat ratio

By combining Equation 15 and Equation 16, the following formula can be derived:

$$P \cdot \left(\frac{P}{R \cdot T}\right)^{-\gamma} = constant \Leftrightarrow P \cdot \left(\frac{R \cdot T}{P}\right)^{\gamma} = constant$$
$$P \cdot (R \cdot T)^{\gamma} \cdot P^{-\gamma} = constant \Leftrightarrow (R \cdot T)^{\gamma} \cdot P^{1-\gamma} = constant$$

Because R is a constant, it can be absorbed in the general constant, so:

$$P^{1-\gamma} \cdot (R \cdot T)^{\gamma} = constant \Leftrightarrow P^{1-\gamma} \cdot T^{\gamma} = constant$$
$$P_0^{1-\gamma} \cdot T_0^{\gamma} = P^{1-\gamma} \cdot T^{\gamma} \Leftrightarrow$$

By dividing both sides by $P_1^{1-\gamma}$:

$$T_1^{\gamma} = \frac{P_0^{1-\gamma} \cdot T_0^{\gamma}}{P_1^{1-\gamma}} = T_0^{\gamma} \cdot \left(\frac{P_0}{P_1}\right)^{1-\gamma} = T_0^{\gamma} \cdot \left(\frac{P_1}{P_0}\right)^{\gamma-1}$$

Then, by taking the γ-th root of the entire expression, this gives the isentropic temperature-pressure relation:

$$T_1 = T_0 \left(\frac{P_1}{P_0}\right)^{\frac{\gamma-1}{\gamma}} \tag{17}$$

By combining the Equation 14 and 17, this gives a first isentropic flow velocity formula, using the specific heat at constant pressure (Equation 18):

$$v^2 = 2 \cdot C_p \cdot (T_0 - T)$$
$$v^2 = 2 \cdot C_p \cdot \left(T_0 - T_0 \left(\frac{P}{P_0}\right)^{\frac{\gamma-1}{\gamma}}\right)$$
$$v^2 = 2 \cdot C_p \cdot T_0 \cdot \left(1 - \left(\frac{P}{P_0}\right)^{\frac{\gamma-1}{\gamma}}\right)$$

$$v = \sqrt{2 \cdot C_p \cdot T_0 \cdot \left[1 - \left(\frac{P}{P_0}\right)^{\frac{\gamma-1}{\gamma}}\right]} \tag{18}$$

Where:

v [m/s]: velocity of the gas

C_p [J/(kg · K)]: specific heat at constant pressure

T_0 [K]: temperature before the nuzzle

P [Pa]: pressure after the nuzzle

P_0 [Pa]: pressure before the nuzzle

Since it is more convenient to use the specific heat ratio instead of the specific heat at constant pressure, the C_p can be expressed as a function of γ and R:

$$\begin{aligned}
\gamma = \frac{C_p}{C_v} &\Leftrightarrow C_p = \gamma \cdot C_v \\
R = C_p - C_v &\Leftrightarrow C_v = C_p - R \\
C_p = \gamma \cdot (C_p - R) &\Leftrightarrow C_p = \gamma \cdot C_p - \gamma \cdot R \\
\gamma \cdot C_p - C_p = \gamma \cdot R &\Leftrightarrow C_p \cdot (\gamma - 1) = \gamma \cdot R
\end{aligned}$$

$$C_p = \frac{\gamma \cdot R}{\gamma - 1} \tag{19}$$

By combining the Equation 18 and 19, this gives the isentropic flow velocity formula using the specific heat ratio (Equation 20):

$$v = \sqrt{\frac{2 \cdot \gamma \cdot R \cdot T_0}{\gamma - 1} \cdot \left[1 - \left(\frac{P}{P_0}\right)^{\frac{\gamma-1}{\gamma}}\right]} \tag{20}$$

Where R is the specific gas constant, which can be calculated from the universal gas constant and the molar mass of the gas:

$$R_{specific} = \frac{R_{universal}}{M} \tag{21}$$

Property	**Helium**	**Neon**	**Argon**	**Krypton**	**Xenon**	**Nitrogen**
Specific gas constant R ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	$2080^{a)}$	$412^{a)}$	$208^{a)}$	$99.2^{a)}$	$63.3^{a)}$	$297^{a)}$
Specif. heat ratio av. γ	$1.67^{b)}$	$1.67^{b)}$	$1.69^{b)}$	$1.73^{b)}$	$1.83^{b)}$	$1.4^{b)}$

Table 3: Summary table of gas properties for velocity calculations. a) The respective specific gas constants have been calculated from Equation 21 and the molar masses shown in Table 1, b) The respective specific heat ratio values are taken from the average of all those given in Table 1, the aim being to provide an estimation for different pressures and temperatures with a simplified approach.

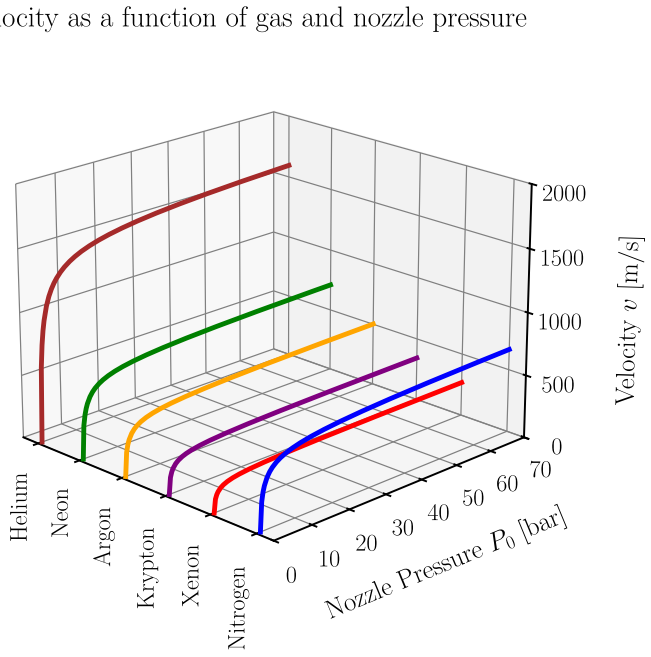

Figure 11: .
Plot of the velocity of different gases at the nozzle outlet as a function of the pressure ratio P/P_0 for a given temperature before the nozzle. Here $\mathrm{T_0}$ = 300 K and P = 1 bar. The velocity is calculated using the isentropic flow velocity formula (Equation 20) and the gas properties from Table 3.

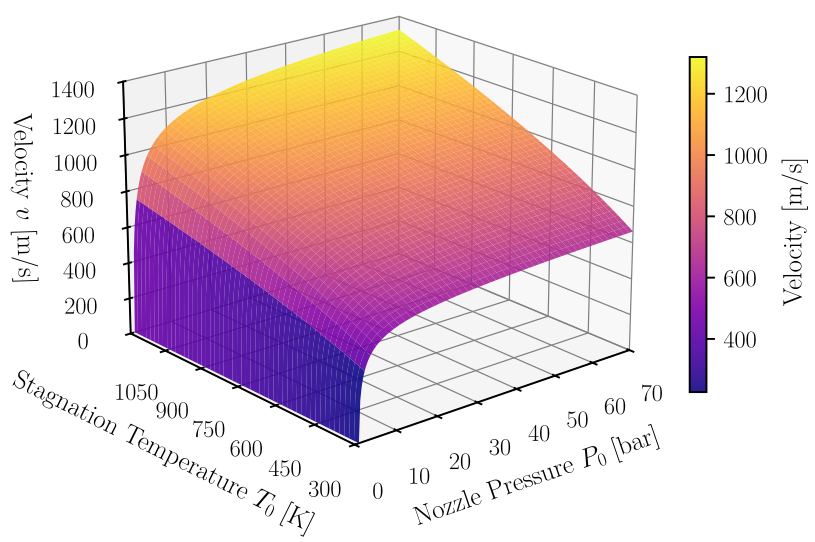

Figure 12: Velocity of Nitrogen as a function of nozzle pressure and nozzle temperature.

With the values relatives to the DTU atomizer, here are the different parameters:

- $\frac{P}{P_0} = \frac{1}{15} = 0.067$
- $\gamma = 1.4$
- $\mathrm{T}_0 = 293$ K
- $R_{nitrogen} = \frac{R_{universel}}{M_{nitrogen}} = \frac{8.314}{0.028} = 297$ J/(kg · K)

By using the isentropic flow velocity formula (Equation 20), the velocity of nitrogen at the nozzle outlet can be estimated around 573 m/s. Mach1 being 348.9m/s according to the data from Figure 9, the Mach number would be then 1.64.

A physical limit imposes a maximum velocity of the gas before the nozzle outlet, which is the speed of sound in the gas. Just after the nozzle outlet, the gas pressure is 50 times lower than the supply pressure, which means that the gas is expanding and accelerating. Due to the turbulence, the gas velocity quickly decrease after reaching its maximum value. The location of this maximum value can be estimated with the Equation 22 [48, 49, 50]:

$$L = k \cdot D \cdot \sqrt{\frac{P_0}{P}} \tag{22}$$

Where:

- D [mm]: diameter of the nozzle
- P_0 [Pa]: supply pressure
- P [Pa]: pressure at the nozzle outlet

The maximum gas speed can be estimated at around 4 mm from the nozzle output, which is a shorter distance than what the "before 2026" nozzle configurations (about 53 mm) allow, taking:

- k = 0.67 [50]
- $D = 1.5$
- $P_0 = 15$
- $P = 1$

In the context of gas atomization, if the goal is to focus on nozzle geometry, an idea can be to start with Nitrogen since it is not expansive and easy to get, knowing that it is not the best option. At some point, gas parameters will have to be considered for:

- cross checking the simulation, maybe exploring optimized mixes of different gases
- gathering the best known parameters to get one batch of good quality powder
- if the chosen gas is not Nitrogen, we could use it during strategic atomization experiment, and go back on nitrogen the rest of the time

≻ Increase the temperature of the gas?

≻ Create a circular flow in the chamber so that the droplet takes all the same direction = less satellites?

4.2.4 Nozzle geometry and flow dynamics

The gas nozzle geometry is one of the most important parameters governing flow dynamics and, consequently, the powder quality. It controls how the gas flows around the molten metal and how efficiently the two phases interact. In addition to geometry, the gas-to-metal ratio (GMR) is another key parameter that describes the overall amount of gas energy available compared to the metal flow, and therefore also strongly affects the atomization process. It is defined as the ratio between the gas mass flow rate and the molten metal mass flow rate:

$$GMR = \frac{\dot{m}_{gas}}{\dot{m}_{metal}} = \frac{\rho_{gas} \cdot A_{gas} \cdot v_{gas}}{\rho_{metal} \cdot A_{metal} \cdot v_{metal}} \tag{23}$$

Where:

- $\dot{m}$ [kg/s]: mass flow
- ρ [kg/m^3]: density
- A [m^2]: area
- v [m/s]: velocity

The GMR represents the amount of kinetic energy available in the gas phase to fragment the liquid metal stream. Increasing the GMR is typically associated with a reduction in particle size. This trend is clearly reported in the Handbook of Non-Ferrous Metal Powders by Handbook of Non-Ferrous Metal Powders [19], where three different batches of powder produced with increasing GMR values (1.77, 2.48, and 4.06) show a consistent decrease in median particle size, where d50 decreases to from 68 µm, 63.5 µm, 37 µm respectively, despite being produced under slightly different conditions. Although these data originate from separate batches, they illustrate a robust and widely observed inverse relationship between GMR and particle size. The dominant role of GMR

has also been highlighted by Cacace ET al. [51], who compared two atomization conditions using different nozzle designs (a standard configuration and an enhanced nozzle). Their results indicate that, despite geometric differences, the influence of GMR remains predominant: an increase of approximately 73% in GMR led to an average increase of about 21% in fine powder yield. This suggests that, while nozzle geometry strongly affects flow structure and local interaction mechanisms, the overall energy input controlled by GMR can act as a governing parameter in determining atomization performance.

Nevertheless, GMR cannot be considered independently from nozzle design, as the efficiency with which gas energy is transferred to the molten metal depends strongly on the flow structure imposed by the nozzle. Consequently, similar GMR values may lead to significantly different atomization outcomes depending on the gas nozzle configuration and operating regime.

Zerwas et al. [52] compared a discrete jet nozzle (with multiple cylindrical orifices) to a convergent–divergent annular slit nozzle for atomizing AISI H13 steel. Despite differences in tower height, secondary breakup was shown to occur very close to the nozzle (< 0.15 m), suggesting that chamber size did not significantly affect the formation of fine particles. Instead, the dominant differences arise from nozzle design and the resulting gas flow characteristics. The annular slit nozzle operates under supersonic conditions (Ma ≈ 1.48), whereas the discrete jet nozzle reaches sonic conditions (Ma ≈ 1). In addition, the two configurations operate over different gas-to-melt ratio ranges (0.3–2.87 for the discrete jet and 1.52–2.96 for the annular nozzle). Since particle size reduction is governed by primary and secondary breakup mechanisms induced by gas–melt interaction, these differences in flow conditions are expected to influence the resulting particle size distribution. As a result, the annular slit nozzle produces less flakes, smaller particles and a better yield for printable powder (especially for LPBF). However, it may require a higher gas consumption especially at high pressure and a lower production rate than the discrete jet nozzle.

Neikov Et al. [19] describe a two stages gas nozzle for water atomization illustrated in Figure 13. The first stage has radial jets, while the second has tangential jets. The introduction of tangential jets generates a swirling motion of the fluid, which significantly influences the atomization process. Compared to a conventional conical jet configuration, the swirl component enhances the breakup of the liquid metal stream, leading to finer particles. In this paper, the use of swirl water jets allows the production of bronze (Cu–10 mass% Sn) powder with a median particle size of 7.5 µm and an apparent density of 2.8 g/cm^3 at a water pressure of 83.3 MPa. Furthermore, the configuration of the jet, particularly the apex angle (γ), plays a crucial role in determining particle size. For conventional conical jets (no swirl), the median particle diameter (d_{50}) decreases with decreasing γ down to about 0.35 rad, below which d_{50} increases abruptly. In contrast, for swirl jet configurations (swirl angle $\theta = 0.2$ rad), d_{50} continues to decrease as γ decreases. In the range $0.35 \leq \gamma \leq 0.87$ rad, the particle size obtained with swirl jets is approximately one order of magnitude smaller than that produced by conventional conical jets. Additionally, powders produced with the conventional configuration exhibit a wide particle size distribution and include many coarse particles with irregular or droplet-like shapes. As γ increases up to 0.87 rad, the particle shape becomes increasingly irregular. In contrast, powders produced using swirl jets consist predominantly of spherical particles with fewer irregular aggregates, and the particle size decreases consistently with decreasing γ.

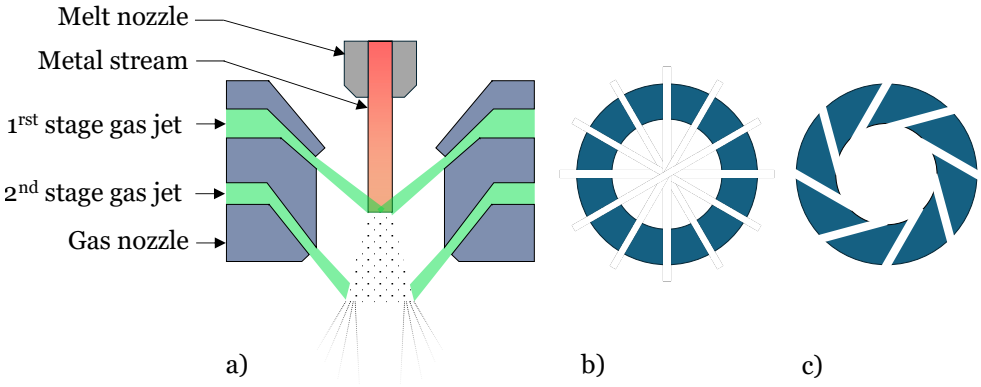

Figure 13: a) Water double stage nozzle. b) Upper stage nozzle with axial jets. c) Lower stage nozzle with tangential jets. Adapted from [19]

Two other nozzle concepts are described for gas atomization. The first one illustrated in Figure 14 a) allows a regime where a film is formed along the end of the melt nozzle, resulting to fine powder production. In addition, it creates a suction effect that prevents from reverse flow towards the melt nozzle. The second one illustrated in Figure 14 b) breaks the melt stream with a friction effect between gaz and metal. By keeping the stream with a small diameter, the powder obtained can be very fine.

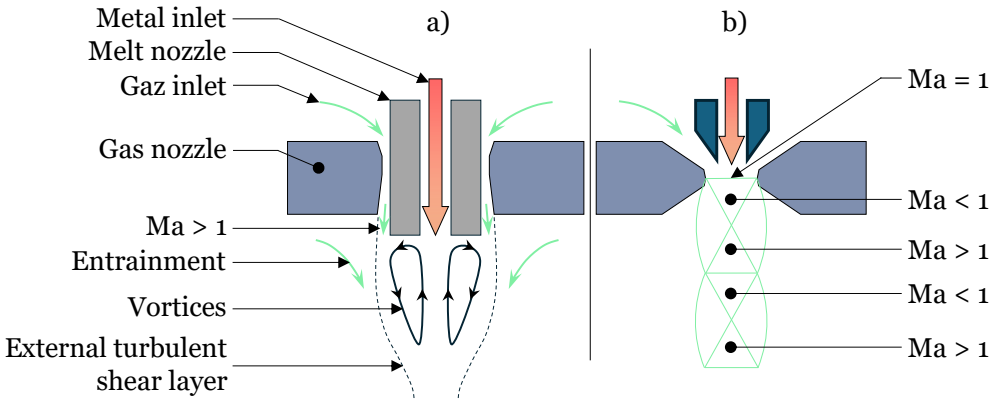

Figure 14: a) Prefilm nozzle. b) Nanoval nozzle. Ma is the Mach number. Adapted from [19, 51].

Choi et al. [53] investigate the atomization of a liquid (water) column subjected to an annular dual-nozzle gas (air) jet configuration, combining both experimental visualization and numerical simulation. Their study identifies four distinct flow regimes governed primarily by the relative momentum flux between the upper and lower gas jets, as well as the liquid flow rate: a confined column regime, a backflow-dominated regime, a bulk atomization regime, and a fine droplet atomization regime. Using high-speed imaging and PIV measurements, they demonstrate that the upper jet primarily governs the initial interaction with the liquid column, including the onset of surface destabilization and the control of the primary deformation location, while also contributing to the suppression of backflow near the injection region. In contrast, the lower jet plays a dominant role in driving the subsequent breakup process and governing droplet formation dynamics.

≻ gas speed / distance out of nozzle Niels PhD p16-17

CAUTION: when designing different nozzle with different apex angle, the impact height will change as well as the distance between gas outlet and impact. To modify only one parameter (the angle), the location and the geometry of the nozzle have to be adapted to keep the other same parameters.

Mates Et al. [54] compare different converging-only nozzle designs (annular slit and discrete jets). The preferred one is the annular slit. Among the discrete jets, the best one is the one that has the smallest jet ring diameters.

Le Et al. [55] reported that the relative position of the gas nozzle and melt delivery tube affects the pressure at the tube tip and the resulting aspiration conditions, thereby influencing the atomization behavior

Ting et al. [56] investigates how gas pressure influences molten metal atomization (Ni-based alloy in that case), with particular focus on the role of wake-closure pressure (WCP). Contrary to common expectations, increasing aspiration pressure beyond the WCP does not increase the molten metal flow rate. Instead, operating above this threshold actually reduces the melt flow. This finding indicates that atomization performance cannot be directly predicted based on aspiration pressure alone. Despite the reduced flow rate, operating slightly above the WCP produces significantly finer powder. The study shows that increasing the pressure by only 0.014 MPa (2 psi) above WCP improves the yield of fine particles by approximately 42%, compared to operating below WCP. However, this improvement comes with a higher gas consumption.

The authors propose a gas-dynamic model in which wake closure reduces the size and strength of the recirculation zone behind the gas flow. This change in flow structure leads to the measured deep aspiration pressures. In addition, a pulsating atomization model is introduced. According to this model, the system does not remain in a steady wake-closed state during operation but instead fluctuates between open and closed wake conditions. The wake is therefore not permanently closed before melt entry. Indeed, the gas flow tends to form a closed wake structure, but the incoming melt disrupts it, temporarily opening the wake, after which it re-forms again. This repeating opening and closing creates a pulsating behavior during atomization.

This pulsating behavior affects how the molten metal breaks up. Above WCP, the process promotes the formation of a thin liquid film, which is more easily fragmented into fine droplets. Below WCP, the melt behaves more like a continuous stream that breaks apart mainly through collisions, similar to free-fall atomization. This study provides a clear explanation for why operating just above the wake-closure pressure leads to finer powders, even when the pressure difference compared to below-WCP conditions is very small. It also highlights the importance of dynamic flow behavior in atomization and suggests that pulsation plays a key role in determining powder characteristics.

Overall, nozzle geometry plays a major role in controlling atomization and the resulting powder properties because it directly affects the gas flow and its interaction with the liquid. Across the different designs, a common trend is observed: nozzle configurations that increase shear, trigger early destabilization of the liquid, and limit undesired effects such as backflow tend to produce finer and more spherical particles. Multi-stage and annular nozzles are particularly effective because they separate functions: the first stage controls the initial interaction and stabilizes the flow, while the second stage enhances breakup and droplet formation. In addition, introducing swirl or film-forming mechanisms further improves atomization by increasing instabilities at the liquid surface. Converging–diverging nozzle geometries, which enable high-speed or even supersonic gas flows, also appears as a promising direction to further enhance atomization efficiency and deserve deeper investigation. However, these benefits often come with higher gas consumption or lower production rates, showing a trade-off between powder quality and process efficiency. Therefore, optimizing nozzle geometry requires balancing these effects rather than modifying a single parameter. Furthermore, studies on WCP highlight that atomization behavior is not only governed by

nozzle geometry but also by distinct flow regime transitions within the same geometry. Near and above WCP, small changes in gas pressure can significantly alter the flow structure, leading to a pulsating atomization regime that promotes melt film formation and results in finer particle sizes. This demonstrates that even within a fixed nozzle design, the internal gas dynamics and operating regime strongly influence atomization efficiency and powder characteristics.

[57]

- Increasing the pressure ratio from 14 to 55 significantly increases the supersonic jet length (approximately $\times 2$ to $\times 4$).
- The maximum dynamic pressure also increases, but more moderately.
- The two types of nozzles (convergent and converging-diverging) show relatively similar behavior at high pressure.
- At a low pressure ratio (about 14), however, the converging-diverging nozzle produces a significantly longer supersonic jet.

[58]

- Atomization begins near the nozzle exit, but this is not where most of the fine particles are formed.
- The large droplets formed initially are subsequently broken into smaller particles during a second stage called secondary breakup.
- Increasing the stagnation pressure results in a longer supersonic region, which promotes secondary breakup and therefore produces a larger amount of fine particles.
- The convergent and converging-diverging nozzles produce fairly similar particle size distributions under most of the conditions studied. Therefore, a converging-diverging nozzle does not necessarily guarantee a finer powder.

- without swirl, do not decrease apex angle under 20 deg
- with swirl 11.5 deg, apex angle can be decreased until 5 deg.
- Annular nozzle + swirl = bring more radial homogeneity
- Laval nozzle to be integrated
- film along the melt nozzle is good, as a first stage gaz nozzle, to guide the melt, bring a first aspiration
- Operating just above the Closed-wake pressure to create oscillating regime

5 Powder characterization

≻ Go to Daniel PhD many info about that

5.0.1 Sphericity and size

5.0.2 Porosity

5.0.3 Oxydation

5.0.4 Ductility and hardness

5.0.5 Impurities and reactivity

5.0.6 Particle size distribution

5.0.7 Tap density, apparent density, compressibility, green strength, flow properties

A simple way to characterize the flowability of powder is to measure the Hausner ratio, which is the ratio of the tap density to the apparent density [2].

The powder flowability is a critical parameter in additive manufacturing, reducing internal stresses, increasing dimensional and shape accuracy and reducing the surface roughness of the printed object. [27].

6 Experiments

6.1 Alumina casting

Alumina powder is used to cast the customized melt nozzle and the inner layer of the crucible. Indeed, the crucible is made of graphite and shouldn't be in contact with the melt. The alumina powder is milled to obtained very thin particles. By mixing it with the right amount of water, this results to a non-Newtonian paste which will be poured in the proper mold. The mold is usually 3D printed with PLA or PETG by fusion deposition modeling. A layer of Vaseline is applied before the cast to make the demolding process easier. In order to minimize the cracks when drying, the amount of water has to be minimized as much as possible, which also make the pouring more difficult. With a mechanical vibrating support beneath the mold, the paste can be made even more viscous, to the point where it would not be able to flow into the mold without the assistance of vibration. Indeed, the strange properties of the obtained paste make it much more fluid when it is vibrated, and come back to solid past when the vibration is stopped.

For the crucible inner layer, the about 2 kg of alumina powder is used, with about 800 ml of water.

6.2 Metal atomization

6.2.1 Optical thermal sensor set-up

Tools:

- furnace
- infrared optical sensor kit, with power-supply, sensor, optical fiber, signal processing unit and USB data converter
- computer

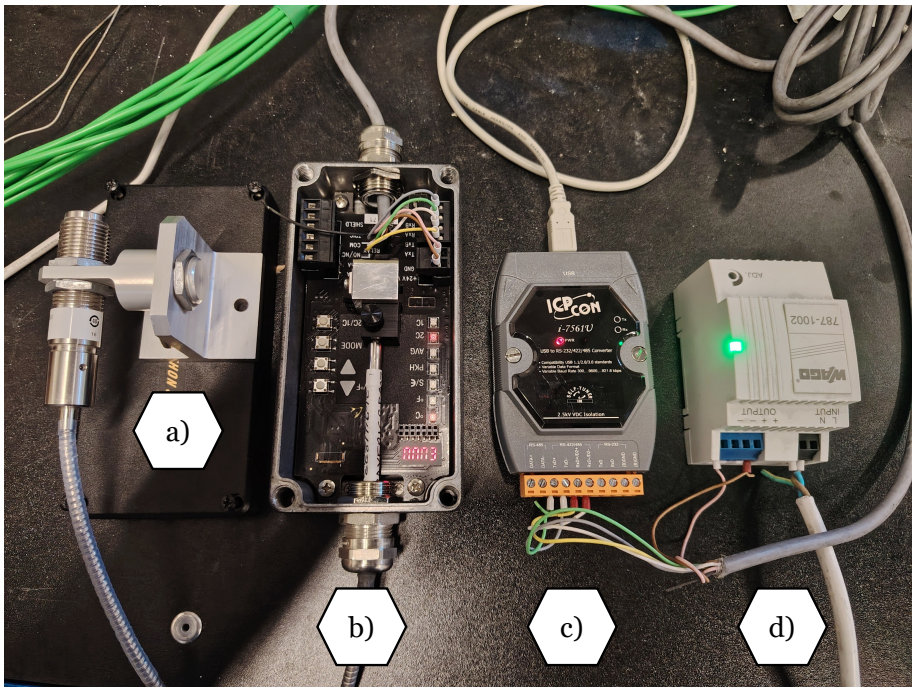

Figure 15: Set up of the optical temperature sensor. a) sensor, b) signal processing unit, c) USB data converter, d) power supply.

The "USB-to-RS232/422/485 converter" can be used only with the proper driver USB-2514_tM-7561_I-756xU. The sensor itself is accessible from the DataTemp Multidrop software.

The signal processing unit has 4 modes of operation:

- **1C (One Color-Mode)**: Standard infrared temperature measurement using a single wavelength. Good for stable emissivity conditions. Use when surface emissivity is known and stable.
- **2C (Two-Color / Ratio Mode)**: Also called ratio pyrometry, uses two wavelengths. Less sensitive to emissivity changes, compensates for dust, smoke, partial obstruction. Use when measuring through dirty optics, target emissivity changes, very hot metals. This mode is the most suitable for our application.
- **AVG (Average)**: Outputs averaged temperature over a time window, smooths signal fluctuations. Use when signal is noisy, process needs stable control input.
- **PKH (Peak Hold)**: Captures and holds the highest detected temperature, useful for fast-moving or small targets. Use when measuring moving objects, induction heating, small hot spots.

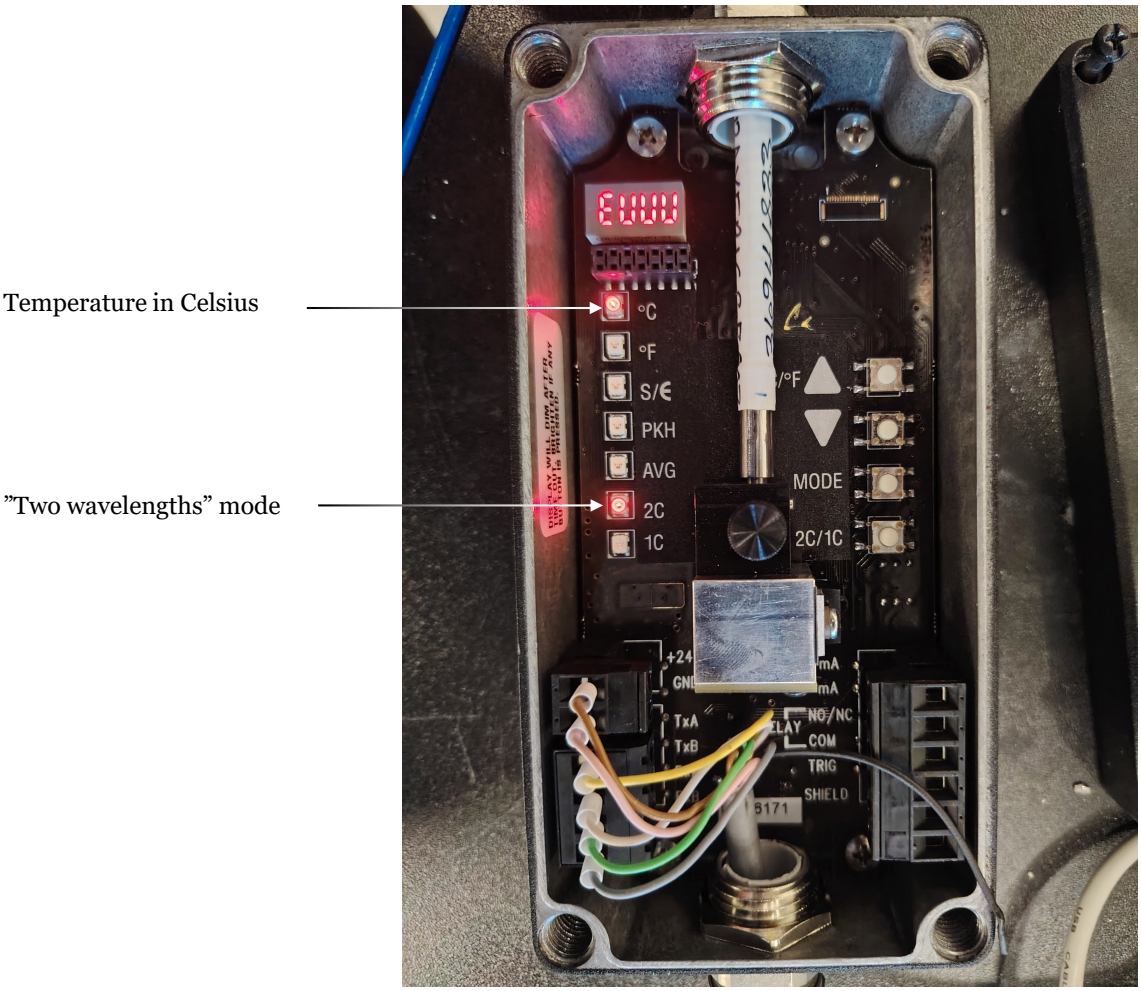

Figure 16: Signal processing unit of the optical temperature sensor kit.

Figure 17: Furnace used to test the optical thermal sensor.

The furnace temperature has been set initially at 1050°C. When the door was opened, the inner temperature dropped down so fast under the lower detection limit of the sensor that it couldn't measure anything or during a too short period of time. Then the furnace temperature has been raised to 1100°C, (which is the maximum temperature) and the sensor was able to measure the temperature for a few seconds before it dropped down again.

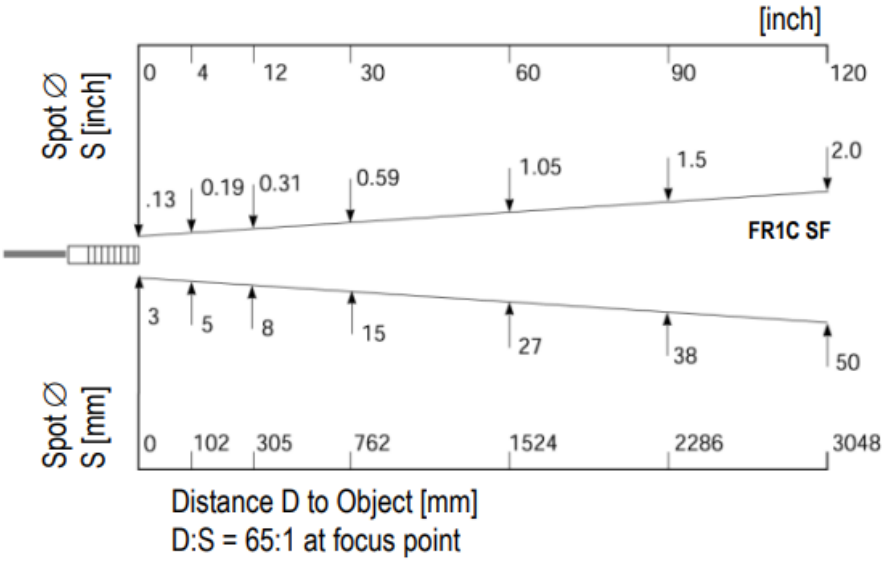

Figure 18: Standard focus spot size chart for the FR1C SF model [59]. The distance between the sensor and the target is estimated to be around 600 mm, which corresponds to a spot size smaller than 15 mm diameter.

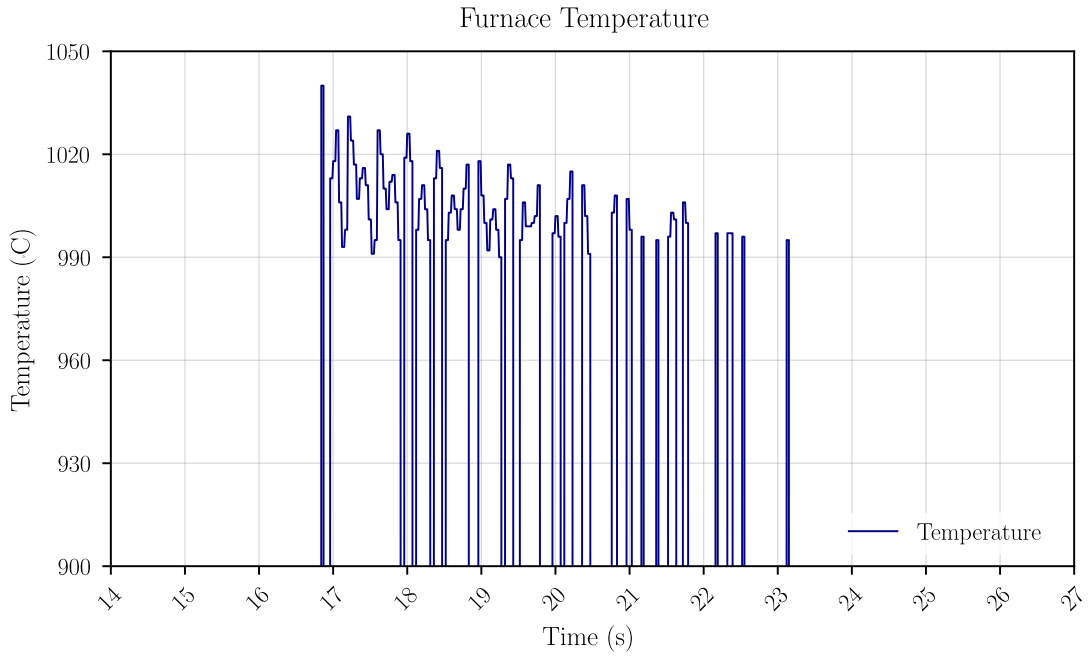

Figure 19: Plot of the temperature measured by the optical sensor when the furnace door is opened.

6.2.2 Atomization setup

The gas atomization system of DTU was located at first in the Risø National Laboratory, and it was moved to DTU during Daniel Cárdenas del Río's PhD [60].The melting chamber contains the crucible, the water cooled induction coil and the gas nozzle. In addition to the original thermocouples an optical thermal sensor [59] is located outside the melting chamber and looks through a dedicated window to measure the temperature of the melt. The part below the melting chamber is the atomization chamber where the molten metal is atomized and and the particles solidify. The gas brings the atomized particles to the bottom of the chamber where they are separated from the gas and collected in the powder container (see Figure 21). The gas is then evacuated in the exhaust system going though two water filters which make sure that no powder particle are still in the gas when it is released to the atmosphere.

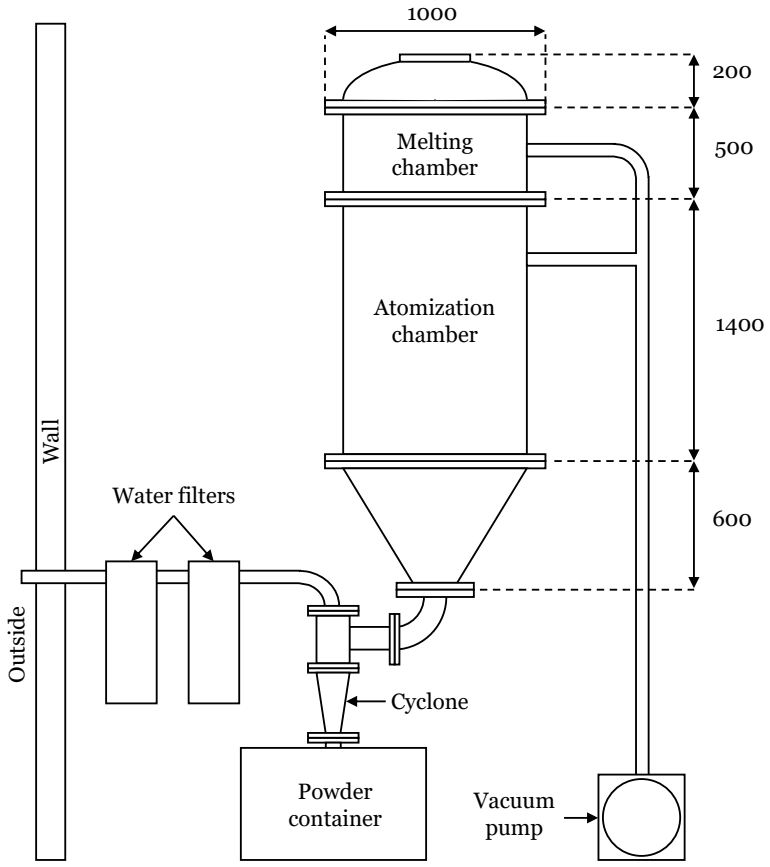

Figure 20: Main components of the atomizer. Measures are in mm.

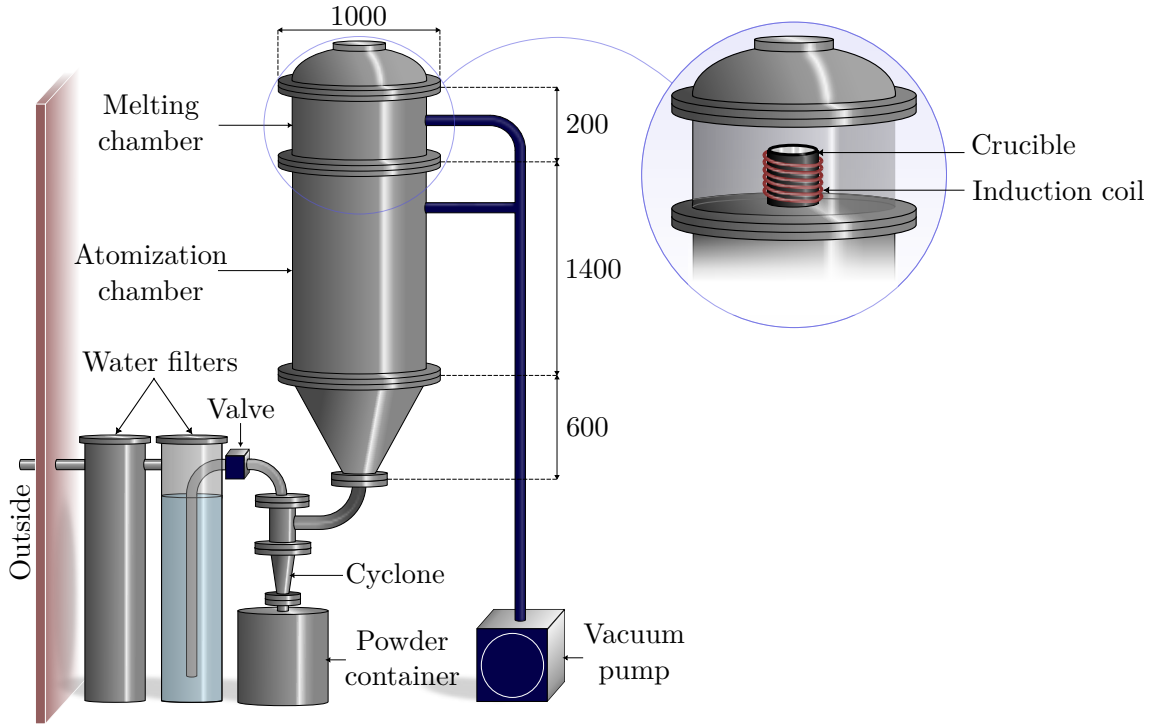

Figure 21: Main components of the atomizer. Measures are in mm.

6.2.3 Atomization 0

The atomizer has to be prepared in advance for the first trial of gas atomization. The temperature of the melt has to be measured during the process, and the optical thermal sensor kit is used for this purpose. The first task is to set up the sensor with the right driver and software and testing the response in front of en opened furnace.

6.2.3.1 Tools / method / protocol

6.2.3.1.1 Preparation of the atomizer

2026/03/17

1. Preparing the crucible, check there is no crack. In case of cracks do some cement and fix them. Dry in the oven the proper way.
2. Put the crucible in the melting chamber. Make sure that it sit properly on the spacers.
3. Put the feedstock rods in the crucible. They should stand vertically letting enough space in the center for the stopper rod. They should be able to extend due to the heat without touching the walls pushing on the crucible inner side (risk of breaking the crucible). This corresponds to approximately 10 kg of stainless steel bars.
4. Put some insolation to cover the top of the crucible and the feedstock rods. An aperture should be left to let the stopper rod go through and also to let the optical sensor look the melt.

5. Put the main cover thanks to the crane and screw it properly to compress the O-ring.
6. Put the main flange and screw it properly to compress the O-ring.
7. Put the stopper rod and fasten it. Raise the horizontal support compressing a bit both of the spring and fasten the rod.
8. Put the optical sensor and fasten it. The sensor should be looking through the dedicated window to the melt.
9. Check the all the valves.
10. Start the vacuum pump until reaching 0.074 bar or lower.
11. Close the vacuum valves, start the N2 flow and switch off the vacuum pump. Pressure should raise to atmospheric pressure.
12. Start the vacuum pump again until reaching 0.074 bar or lower (one N2 flush is done).
13. Close the vacuum valves, start the N2 flow and switch off the vacuum pump. Pressure should be just above atmospheric pressure. (N2 pressure should be 0.15 / 0.18 bar)
14. Oxygen percentage should be 0.2 / 0.3 in the N2

    ≻ How to calibrate these sensors?

6.2.3.1.2 Powder measurement

1. Collect the powder in a heat proof container
2. Put it in an oven during 12 hours at 120°C to dry it.
3. Separate the flaxes from the power by choosing the right sieve size.
4. Then use the standard sieves to separate the powder in different size classes and weight them.
5. Clean the sieves and the container.

6.2.3.2 Results

Thanks to the latest improvement of the atomizer, the vacuum rate was very good, reaching 0.074 bar and the oxygen level was around 0.2% in the N2 after the two flushes, once the atomizer was under N2 atmosphere, ready for atomization. A very small amount of oxidation were floating on the melt surface, which shows that the oxygen level was quite low.

Approximately 10 kg of stainless steel bars were put in the crucible as a feedstock. During this atomization1, 3 main problems appeared:

- A leakage between the metal nozzle and the crucible allowed the liquid metal to go in between
- A freeze-off appeared at the end of the atomization
- The cyclone get stuck very early in the atomization, preventing the powder to go in the powder container. Instead, the powder was drained with the gaz in both of the water filters.

After the atomization, the weight of the obtained materials have been weighed:

- The metal freeze attached to the nozzle: ≈ 500g

- Flaxes in the container: ≈ 3040 g
- Powder in the container [125 < x < 63]: 806 g
- Powder in the container [x < 63]: 121 g
- Flaxes and powder in the cyclone: ≈ 400 g

6.2.4 Atomization 1

6.3 Atomization test station

6.3.1 MN pressure with FDM gas nozzles

6.3.1.1 Abstract

The reliability and performance of additive manufacturing processes depend heavily on the quality of the metal powders used as feedstock, with particle size distribution and morphology serving as key quality indicators. Among the techniques used to produce these powders, gas atomization remains one of the most established, and its outcome is highly sensitive to the design of the atomization nozzle and the conditions under which it operates—factors that govern the surrounding gas flow field, its interaction with the melt stream, and the tendency toward backflow or nozzle blockage through freezing. This study examines how variations in nozzle geometry and process parameters influence the pressure at the melt nozzle within a vacuum induction gas atomization (VIGA) system. To enable systematic testing, a modular experimental rig was built using nozzle components produced via fused deposition modeling (FDM), allowing independent adjustment of apex angle, vertical positioning of the nozzle, swirl angle, and inlet gas pressure. Both single-stage and dual-stage nozzle arrangements were evaluated. Findings indicate that apex angle has a pronounced effect on melt nozzle pressure, an effect that becomes more pronounced when the gas nozzle sits below the outlet of the melt nozzle. As the nozzle's vertical position was raised, melt nozzle pressure shifted from negative to positive values, with this crossover occurring in the range of roughly 4–5 mm for the configuration tested. Adding swirl to the gas flow tended to enhance suction and lower the likelihood of backflow. The authors note that additional experimental work and more refined numerical models will be needed to fully characterize the flow behavior involved and to confirm these preliminary trends.

6.3.1.2 Introduction

Since its commercialization in the late 20th century, additive manufacturing (AM) has experienced rapid growth, leading to a significant increase in the demand for high-quality metallic powders [1]. Numerous processes have been developed to produce metallic powders, each offering different levels of productivity, cost, and powder quality, which inherently determine their suitability for specific applications. Among the key powder characteristics, particle size distribution (PSD) and particle sphericity play a central role, as they directly influence powder flowability, packing density, and process stability. In powder bed-based AM processes, such as laser powder bed fusion (LPBF), variations in powder morphology can lead to poor layer uniformity, defect formation, and variability in the mechanical properties of the final components. Consequently, controlling powder characteristics during production is of critical importance.

6.4 Vacuum Induction Gas Melting Atomization (VIGA)

The gas atomization has been introduced in the 1980s by Crucible Materials Corporation [2]. The VIGA is currently the most common process for producing metallic powders for additive manufacturing. The feedstock is melted in a crucible by induction heating, then atomized by high velocity gas jets. The atomization can be done in a vacuum or in an inert gas atmosphere, which allows it to produce high quality powders with low oxygen content and good sphericity. The VIGA process is divided in two categories: the free fall and confined gas atomization. The main difference is the distance between the bottom of the liquid metal nozzle and the gas impact point. On the free fall side, the metal stream is hit after a certain distance (d_1) on Figure 1 a), which can be more than 100 mm under the delivery tube [3]. An important distance between the liquid metal nozzle and the interaction point helps to prevent the backflow and freeze-off risk. However, the metal stream properties is less optimal: the stream becomes larger because of the surface tension, faster due to the gravity, colder because of the heat loss during the free fall, which can lead to a less efficient atomization and a larger powder size distribution.

FIGURE FREE FALL AND CONFINED

The gas nozzle is therefore a key component of the VIGA process. Its geometry determines the structure, velocity, and direction of the gas flow and consequently influences momentum transfer to the molten metal, jet stability, fragmentation mechanisms, particle size distribution, and particle morphology. Conventional nozzle designs can therefore impose a trade-off between efficient melt fragmentation, fine-powder yield, gas consumption, and process stability.

6.4.0.0.1 Literature on gas jet configuration

Instead of tackling the full atomization process and the resulting powder properties as a single problem, this work intentionally isolates the specific failure mode of backflow from the later stages of atomization and powder formation. Attention is therefore concentrated on the flow behavior near the gas and melt nozzles, with the goal of pinpointing which geometric parameters control the pressure and velocity fields responsible for backflow. More specifically, the study seeks to identify nozzle configurations that generate the desired suction effect while minimizing the risk of melt backflow and the resulting freeze-off. This strategy offers a controlled setting in which nozzle geometry can be studied independently of final powder characteristics, and represents a necessary precursor to their eventual practical application. The complex gas recirculation that occurs around the nozzle is a well-known issue in gas atomization and can generally be split into two categories. In the first, backflow moves toward the melt nozzle, where it can cool and solidify the molten metal, clogging the nozzle and potentially blocking the melt stream from forming at all. In the second, backflow travels along the outside of the melt nozzle toward the gas nozzle dies, sweeping already-atomized particles along with it. These particles can then accumulate and block the gas die, disrupting the atomization process and possibly halting it entirely.

Motaman et al. [61] examined, both numerically and experimentally, how the gas jet mismatch angle and stand-off distance influence back-stream flow in close-coupled gas atomization, demonstrating that reduced mismatch angles and lower gas pressures encourage back-streaming, whereas larger angles ($\geq 5°$) or elevated pressures suppress or eliminate it altogether. Their paper also weighs the relative advantages of free-fall versus close-coupled gas atomization: the gas-to-melt interaction is more easily controlled in the free-fall configuration, while close-coupled atomization yields a superior particle size distribution and higher process efficiency. In fact, both the melt stream and gas jet are more manageable in close-coupled setups, and energy losses are lower, which in turn reduces gas consumption. Their experiments relied on single-stage discrete jet nozzles fitted with convergent-divergent dies. When standard dies are used instead—cylindrical discrete jets, for example—the jet expands in an uncontrolled manner, producing shocks and lowering velocity. A convergent-divergent die, by contrast, can be engineered so that ambient pressure is achieved right at the nozzle outlet. Even so, the configuration examined still exhibited some backflow, causing

metal to freeze at the tips of the gas nozzle.

6.4.0.0.2 Literature on Effect of nozzle geometry on powder quality

Nozzle geometry has a direct bearing on the gas flow field and, by extension, on how the high-velocity gas interacts with the molten metal. This interaction shapes both the initial breakup of the melt stream and the further fragmentation of droplets, which makes nozzle design a key factor governing the resulting particle size distribution. Zhang et al. examined how nozzle angle and position affect the flow field, finding that both parameters reshape the recirculation zone and the gas–liquid interaction near the melt delivery tube [62]. Notably, nozzle angle exerted a greater influence on the flow field than nozzle position did.

These flow-field changes carry through to the powder that is ultimately produced. Within the geometry range studied, smaller nozzle angles yielded finer powder, and the combination of a 33° angle with a 10.15 mm nozzle position produced the smallest characteristic particle size. The study also revealed an interaction between atomization pressure and nozzle geometry: raising the pressure initially reduced the median particle diameter D_{50}, with an optimum appearing near 2 MPa. Taken together, these findings show that even modest adjustments to nozzle geometry can substantially change the gas–melt interaction and, consequently, the particle size distribution of the atomized powder.

6.4.0.1 Material and Methods

6.4.0.1.1 Parameters

Figure 22 shows a scheme listing of parameters that can be varied.

- P: gas inlet pressure [bar]
- α: apex angle [deg]
- ω: swirl angle [deg]
- Z: vertical nozzle position [mm]
- d: vertical distance between stage 1 and 2 [mm]
- RJ: inner radius if annular slit die, or radius of the minimum tangent circle if the discrete jet dies [mm]
- θ: difference between outer and inner diameter of the annular slit die, or diameter of the discrete jet dies [mm]

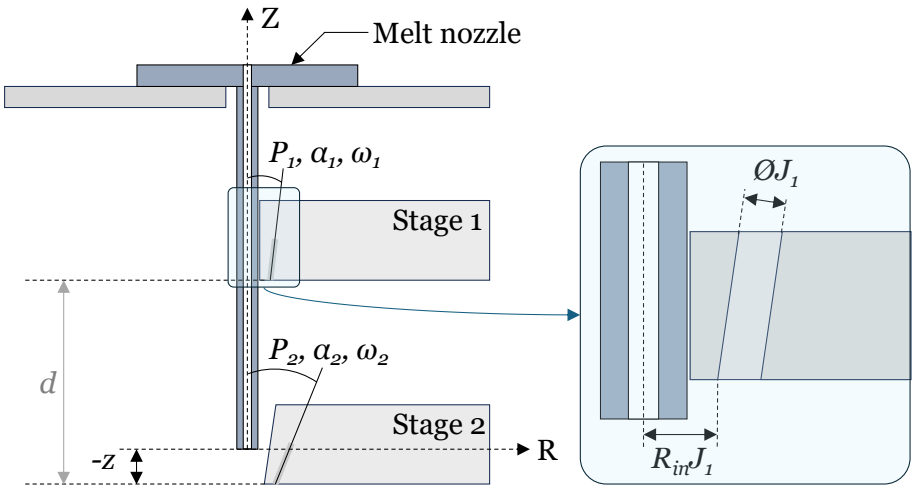

Figure 22: Scheme of the parameters

6.4.0.1.2 Set-up description

The experimental setup illustrated in Figure 23 was designed to allow systematic investigation of the influence of nozzle geometry and operating conditions on the pressure conditions in the melt delivery region. The setup consisted of two independently controlled compressed-air inlets, enabling experiments to be performed using either a single-stage or dual-stage nozzle configuration. For each inlet, the incoming air supply was connected to a flow distributor that divided the system inlet into 20 individual tubes. These tubes transported the air from the supply system to the atomization nozzle, where the 20 individual flow paths were collected and redirected towards a single annular slit outlet. Two Danfoss AKS32 pressure transducers (-1 to 6 bar range) were used with their corresponding NI input module and Flexlogger Software to measure both inlet pressures and the melt nozzle pressure.

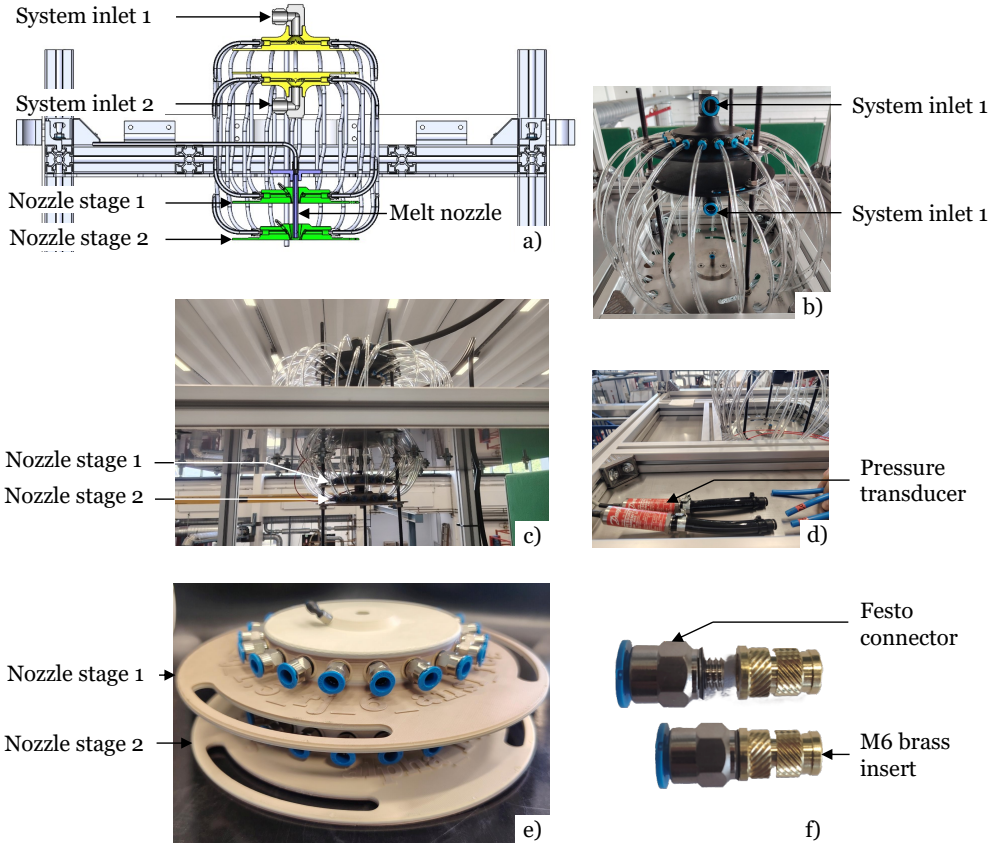

Figure 23: Experimental set-up. a) Cross section of a CAD view presenting the air spreader (yellow) and the customizable air nozzle (green). b) Upper part of the set up showing the inlet1 and inlet2 of the system. c) Lower part of the set-up showing the gas nozzle stage1 and stage2. d) Pressure transducer used to measure the relative pressure in each gas nozzle and in the melt nozzle, located in the center of the gas nozzle.e) Dual stage nozzle printed with FDM. f) Leak-tight connection between the PU tube and the polymer nozzle.

The central feature of the experimental setup is the modular nozzle parametric design. The nozzle components, shown in Figure 23 e) are manufactured by FDM additive manufacturing [63], allowing easy iterative testing at very low cost and enabling rapid, flexible, and time-efficient implementation.

Several nozzle variants were therefore produced with different geometrical parameters. Depending on the investigated configuration, either one or two nozzle stages were installed. In the single-stage configuration, only one of the air circuits and nozzle stages was operated, whereas the dual-stage configuration used both independently supplied stages. This modular approach made it possible to systematically vary parameters, such as the nozzle position and nozzle angle, as illustrated in Figure 22 b), while maintaining the same surrounding experimental equipment.

The distance "d" is kept as a constant as small as possible to fit with the real life constrains: in the atomizer, the melt tube has to be as short as possible to prevent the melt to cool down. As a result, for thermal and physical reasons, melt nozzle and gas nozzle should be as compact as possible. As a result, only "Z" is kept as a vertical shift of both nozzle 1 and 2 with respect to the reference, which is the end of the melt tube.

6.4.0.1.3 Nozzle Configuration naming

Since a large number of configurations are investigated by systematically varying several parameters, a specific naming convention is introduced to uniquely identify each configuration and clearly define the value of every parameter. This notation facilitates the comparison and reproducibility of the different nozzle configurations.

As an example, the configuration name:

"P_1_3_Alpha_3_14_Omega_0_0_Z2_0_DJ_1.5_1.5_RJ_6_7.5_AnSl_Reg"

means:

- it is a dual stage configuration since there are 2 values per variable names
- the pressure in the inlet 1 and 2 is respectively 1 and 3 bar
- Alpha = 3° in the stage 1, and Alpha = 14° in the stage 2
- Omega = 0° in both stages 1 and 2
- $Z_2 = 0$ mm
- DJ = 1.5 mm, in both stages 1 and 2
- RJ = 6 mm in the stage 1, and RJ = 7.5 mm in the stage 2
- both is the nozzle have a regular annular slit die
- when "!" is written instead of a value, it means the value is a variable

6.4.0.2 Experimental results

6.4.0.2.1 Single-stage nozzles

Figure 24 shows that, for $Z_2 \leq 0$, the apex angle (α) has a substantially greater influence on the melt nozzle pressure than the vertical nozzle position (Z_2). This is reflected by the curves being primarily grouped according to line style, representing the different α-values, rather than by colour, representing Z_2. Increasing α shifts the melt nozzle pressure towards more positive values, corresponding to an increased tendency for backflow. For $Z_2 > 0$, however, the trend is reversed. At $Z_2 = 5$ mm, the largest apex angle produces the strongest suction, whereas $\alpha = 0°$ results in positive melt nozzle pressures and therefore an increased tendency for backflow. This change in behavior is likely related to the direct interaction between the gas flow and the melt nozzle when the nozzle stage is positioned above the melt nozzle outlet. This interaction alters the local flow field and may introduce flow separation and increased turbulence, thereby changing the pressure distribution around the melt nozzle.

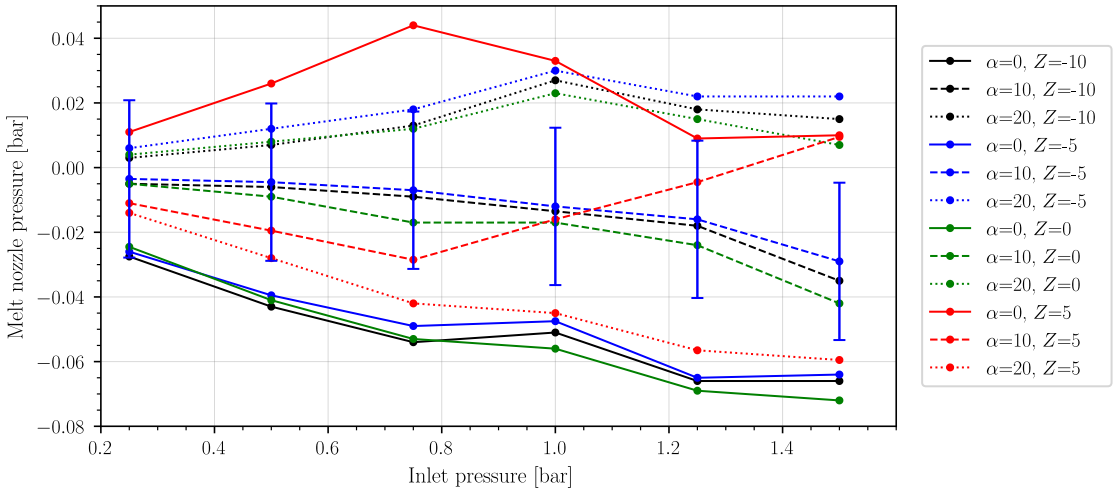

Figure 24: Plot rasing the influence of the "Z" parameter on the melt nozzle pressure. Error bars represent the combined measurement uncertainty, accounting for the pressure sensor accuracy and user reading uncertainty. For clarity, error bars are shown only for [α=10, Z=-5] and are representative of all experimental curves.

Since the results in Figure 24 indicate a pronounced change in behavior between $Z_2 = 0$ mm and $Z_2 = 5$ mm, this region was investigated in greater detail to identify where the transition occurs. Figure 25 shows the melt nozzle pressure as a function of Z_2 for several inlet pressures. For $Z_2 \leq 3$ mm, the melt nozzle pressure remains negative for all investigated inlet pressures, corresponding to suction. As the nozzle is shifted further upward, the melt nozzle pressure increases rapidly, with the transition from negative to positive pressure occurring between approximately $Z_2 = 4.5$ mm. This indicates a distinct change in the local flow behavior as the gas jet begins to interact more strongly with the melt nozzle geometry. The exact transition position depends somewhat on inlet pressure, but the general trend is consistent across the investigated pressure range.

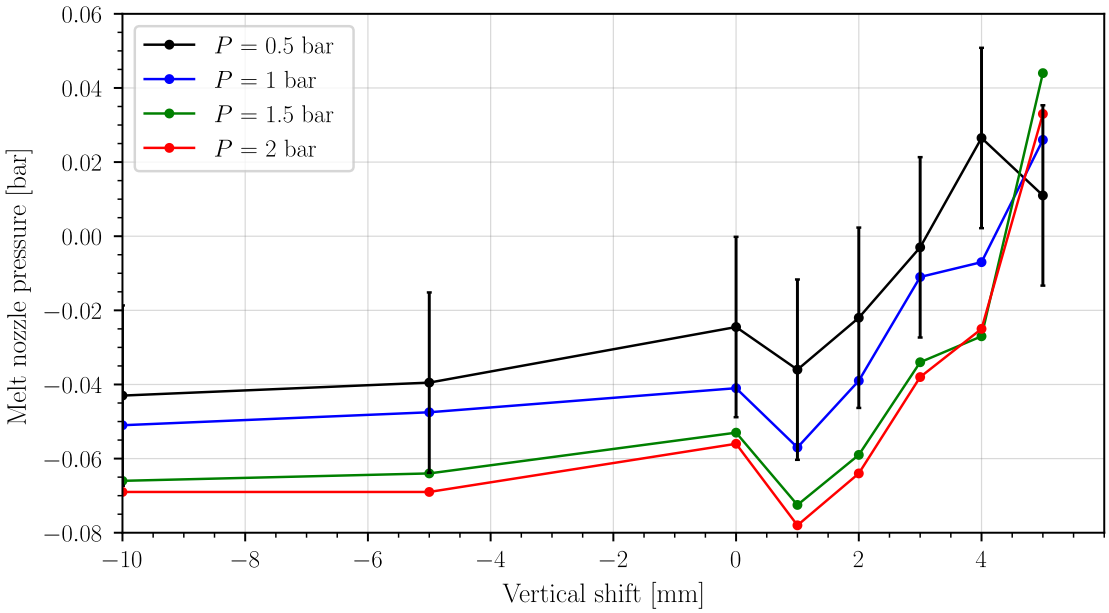

Figure 25: Plot raising the sensitivity of the vertical shift along the melt nozzle, especially in when Z is close to 4 mm. Error bars represent the combined measurement uncertainty, accounting for the pressure sensor accuracy and user reading uncertainty. For clarity, error bars are shown only for [P=0.5 bar] and are representative of all experimental curves.

To evaluate the impact of swirl in a single-stage nozzle, configurations with a swirl angle of $\omega = 45°$ were compared against non-swirled configurations ($\omega = 0°$). At $\alpha = 14$, the $\omega = 45°$ nozzle yields consistently better performance than the $\omega = 0°$ case across the Z_2 positions tested, as illustrated in Figure 26. These results suggest that the helical flow component lowers the pressure within the melt nozzle, thereby strengthening the suction effect. A similar pressure reduction was reported by Yoshimura et al. (2022) [64], lending support to this finding; however, in their case the reduced jet velocity at the nozzle lip led to coarser particle formation rather than the improvement observed here. Given this, a key question going forward is whether the low-pressure region persists once a second atomization stage is introduced in a dual-stage configuration.

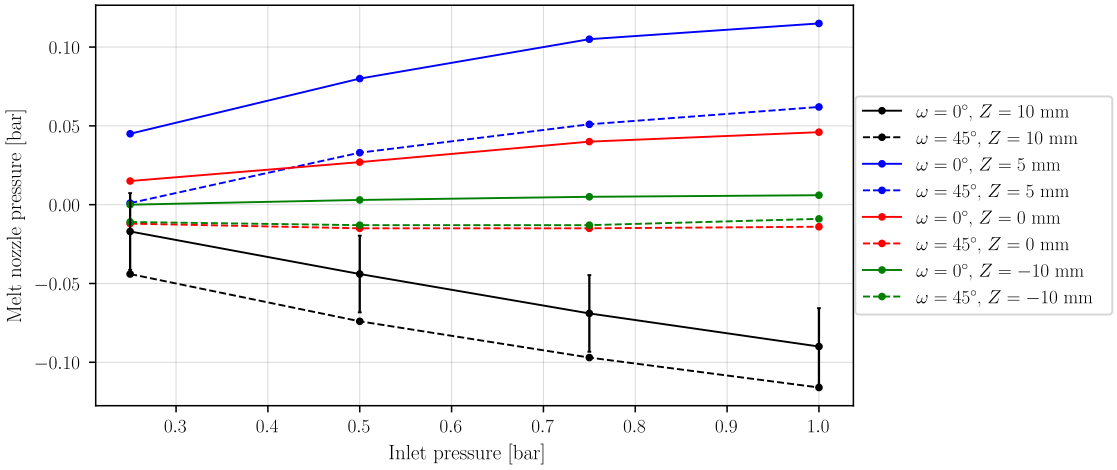

Figure 26: Plot raising the influence of the swirl angle on the melt nozzle pressure. Error bars represent the combined measurement uncertainty, accounting for the pressure sensor accuracy and user reading uncertainty. For clarity, error bars are shown only for [ω=0, Z=10] and are representative of all experimental curves.
Pressure in the melt nozzle as a function of inlet pressure, varying swirl angle (ω) and vertical shift (Z) P_!_Alpha_0_omega_!_Z_!_DJ_1.5_RJ_6_AnSl_Reg

6.4.0.2.2 Dual-stage nozzles

Stage 1 Omega 0 stage 2 Omega 0

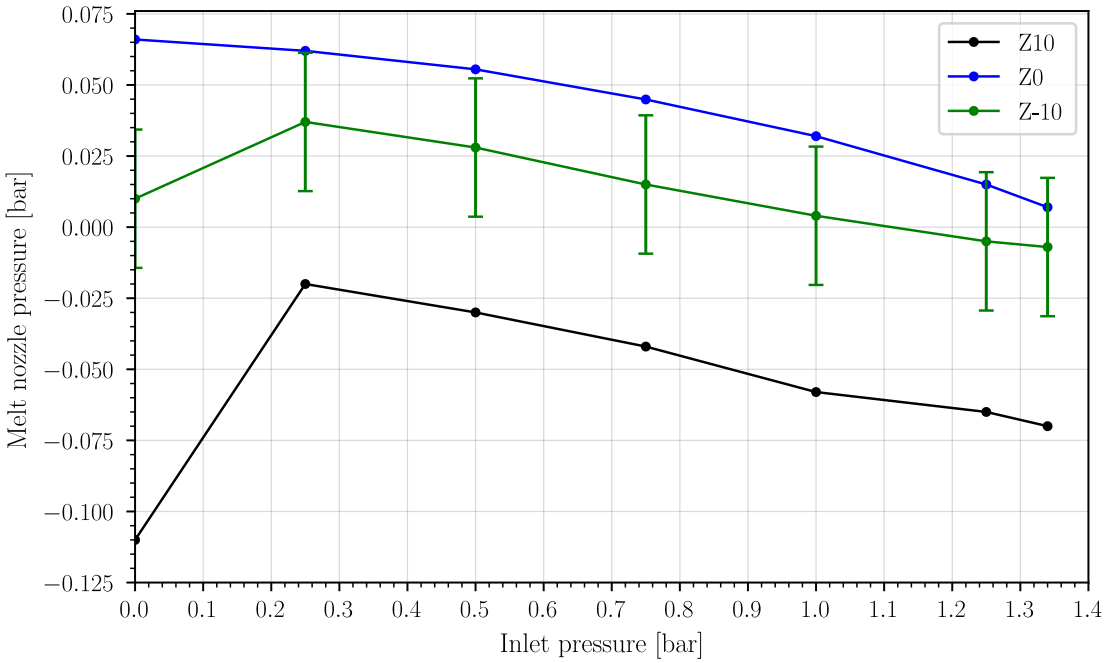

Figure 27: Pressure in the melt nozzle as a function of stage 1 inlet pressure (P_1), keeping stage 2 at 1.07 bar (P_2), varying gas nozzles vertical shift (Z)
P_!_1.07_Alpha_3_14_Omega_0_0_Z_!_DJ_1.5_1.5_RJ_6_7.5_AnSl_Reg

Stage 1 Omega 45 stage 2 Omega 0

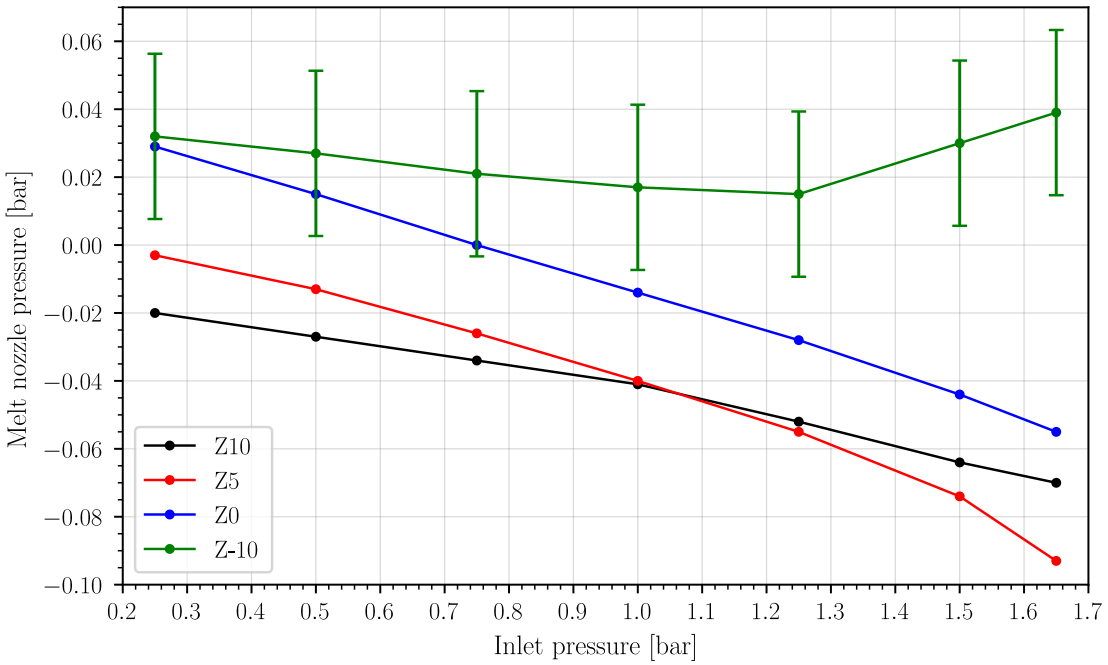

Figure 28: Pressure in the melt nozzle as a function of stage 1 inlet pressure (P_1), keeping stage 2 at 0.72 bar (P_2), varying gas nozzles vertical shift (Z)
P_!_0.72_Alpha_3_14_Omega_45_0_Z_!_DJ_1.5_1.5_RJ_6_7.5_AnSl_Reg

Stage 1 Omega 45 and stage 2 Omega 45

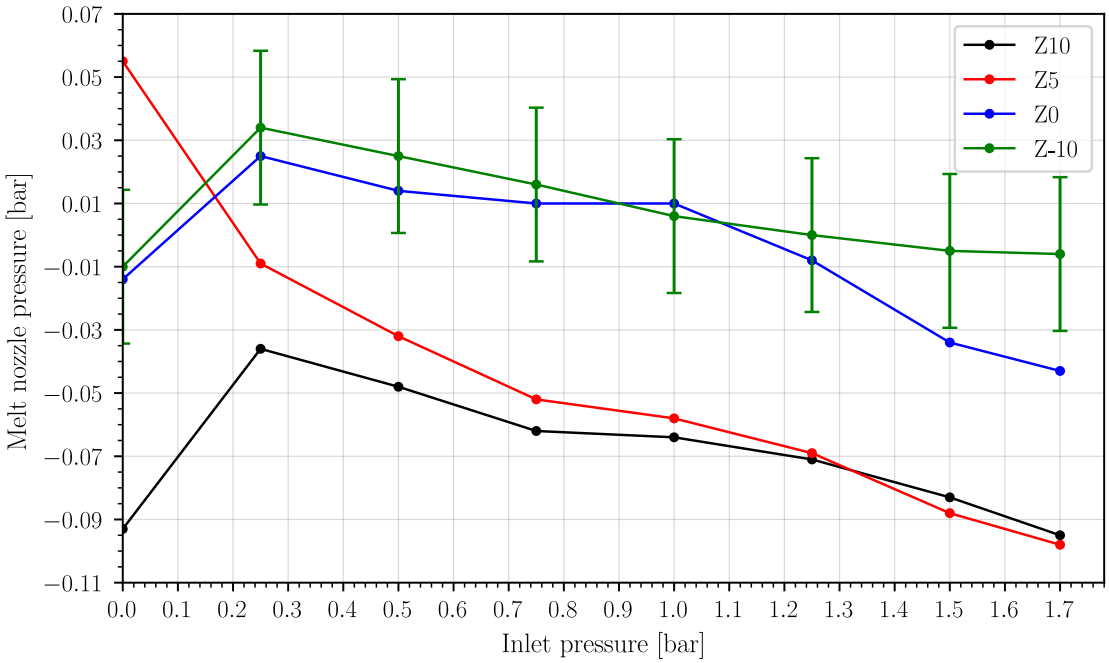

Figure 29: Pressure in the melt nozzle as a function of stage 1 inlet pressure (P_1), keeping stage 2 at 0.72 bar (P_2), varying gas nozzles vertical shift (Z)
P_!_0.72_Alpha_3_14_Omega_45_45_Z_!_DJ_1.5_1.5_RJ_6_7.5_AnSl_Reg

6.4.0.2.3 Error bar

From the data sheet, the accuracy is read to be 0.3% FS (typ) [65]. This accuracy includes hysteresis, repeatability and non-linearity. To get this in bar instead of a percentage, the value is multiplied with the full scale (sensor range), which is from -1 to 6 bar = 7 bar. The value is converted to a standard deviation.

$$\frac{0.3 * 0.01}{\sqrt{3}} * 7 = 0.01212bar \tag{24}$$

Different errors such as thermal sensitivity and supply voltage dependency are also given in the data sheet, but both the temperature and the supply voltage are assumed constant and these terms are therefore negligible.

The error from the noise is found empirically and is found to be 0.001 bar after a low-pass filter with a cut-off value of 0.2 Hz has been applied.

The values are combined with the Residual Sum of Squares method and a coverage factor of 2 is used to put the error in the 95% confidence interval.

$$\sqrt{0.01212^2 + 0.001^2} \times 2 = 0.02433bar \tag{25}$$

Here it is apparent that the accuracy-error given from the datasheet is an order of magnitude larger than the noise-error. So although the noise error have minor fluctuations at different inlet pressures, this has been found to be negligible and the error is regarded as constant.

6.4.0.3 Others info

6.4.0.3.1 Nozzle design ideas

FIRST STAGE

⇒ focus metal stream

⇒ protect from backflow

⇒ create a vacuum effect and first slight breakup

- minimum apex angle
- annular (and swirling, to decrease even more apex angle)
- low pressure
- starting before the melt nozzle end, guided but the melt nozzle. (In the future, the melt nozzle and the gaz nozzle should be integrated in one nozzle, in printed ceramic)

SECOND STAGE

⇒ suck the 1st stage flow and the liquid stream

⇒ secondary breakup

- very small apex angle
- Laval geometry
- annular and swirling
- high pressure, just above the closed-wake pressure

6.4.0.3.2 Printing tests for FDM nozzles

- stage1, only 3 slots with pressure transducer slot, side wall 1.5 mm: test sealing with one PU inlet and two clogged ones, outlet clogged.
- stage1, only 3 slots with pressure transducer slot, side wall 1.5 mm: test sealing with one PU inlet and two clogged ones, outlet open.
- Stage1, full manifold: test sealing with one PU inlet and all the outlet closed.

Results: as expected, small wall thickness brings to more porosity. Choice to go with full of plastic, however, under high pressure, still porous. In the real life, the pressure inside the manifold will be lower since the circuit will be open.

6.4.1 Background oriented schlieren (BOS)

6.4.1.1 Literature on BOS

Schlieren imaging encompasses a broad family of techniques, implemented in many different ways, for visualizing the internal structure of transparent flows characterized by non-uniform refractive index fields. This work concentrates on the Background-Oriented Schlieren (BOS) approach,

the newest variant within this family and one that continues to drive innovation among experimental researchers. This technique was independently introduced in the peer-reviewed literature in the year 2000: Dalziel and colleagues referred to it as "synthetic Schlieren" [66], while Raffel and coworkers coined the term "Background-Oriented Schlieren" [67]. The latter designation has since become the standard terminology within the fluid mechanics community.

Compared to other Schlieren-based methods, BOS offers several benefits, chief among them the minimal experimental requirements: only a camera to photograph a textured background and the background pattern itself are needed. A BOS measurement quantifies how the recorded pattern appears distorted from the camera's viewpoint, a distortion caused by light-ray bending due to refractive-index variations occurring in the region between the camera and the background. While a synthetic random pattern is typically projected or printed for use as the background, naturally occurring textures can serve the same purpose such as landscapes [68], and even the granular intensity variations of band-pass-filtered sunlight at the solar surface [69]. These natural-background applications have notably enabled the visualization of aerodynamic flow fields surrounding helicopters and fighter jets in flight [70, 71, 72].

Several studies have systematically compared the influence of background-pattern properties (including dot shape, color, and grayscale content) on BOS measurement quality [73, 74]. Hu et al. [73] evaluated semi-random dot patterns alongside fractal-like backgrounds generated by diffusion-limited aggregation (dot-DLA), reporting that the conventional dot pattern was the most robust choice for well-defined explosion flows, whereas the dot–DLA pattern performed best at capturing fine-scale turbulent structures. Given its robustness across flow regimes and its comparative simplicity to generate and implement, the dot pattern remains the most widely adopted background in BOS experiments, and it is the pattern retained in the present work.

7 Design and updates

7.1 Re-design from the existing 90's paper drawings

2026/03/12

From the original 2D drawings and from measurements of the atomizer, CAD 3D model has been created for several reasons:

- Allowing a strong base to help the design of the modified parts which have to be adapted to the existing system
- Having an updated status of the atomizer with the implementation of the modified parts and updates
- Having a direct connection and an easy access the dimensions of the environnement for the numerical simulation
- Make a good base for illustration in reports and presentations, help for understanding the system and phenomenon involved in the atomizer.

the integration of the atomizer has been re-designed.

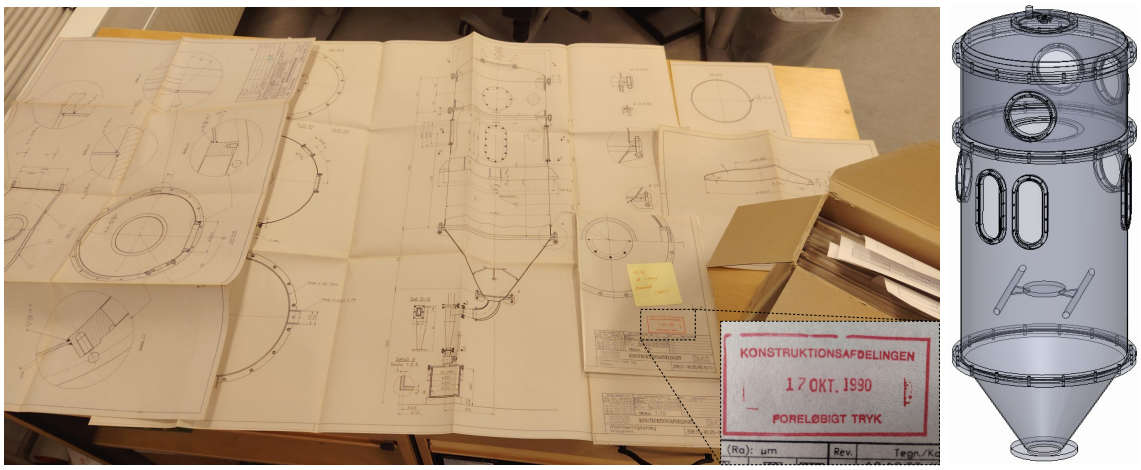

Figure 30: Redesign of the atomizer based on the original drawings from the 90's. On the left, the 5 kg batch of assembly and detailed paper drawings from 1990. On the left, the re-built CAD model from the drawings and the atomizer itself.

7.2 Upper flange and lifting tool

The upper flange of the atomizer weighs around 25 kg and had to be installed and removed by hand on the top of the atomizer. This operation being quite difficult and dangerous, a lifting tool has been designed to be able to be handled by a crane. To unsure that the assembly is well dimensioned, a basic calculation has been done with ANSYS Workbench 2025 R2 [75]. Figure 31 shows the assumptions, boundary conditions and the mesh used for the calculation. The results are presented in Figure 32.

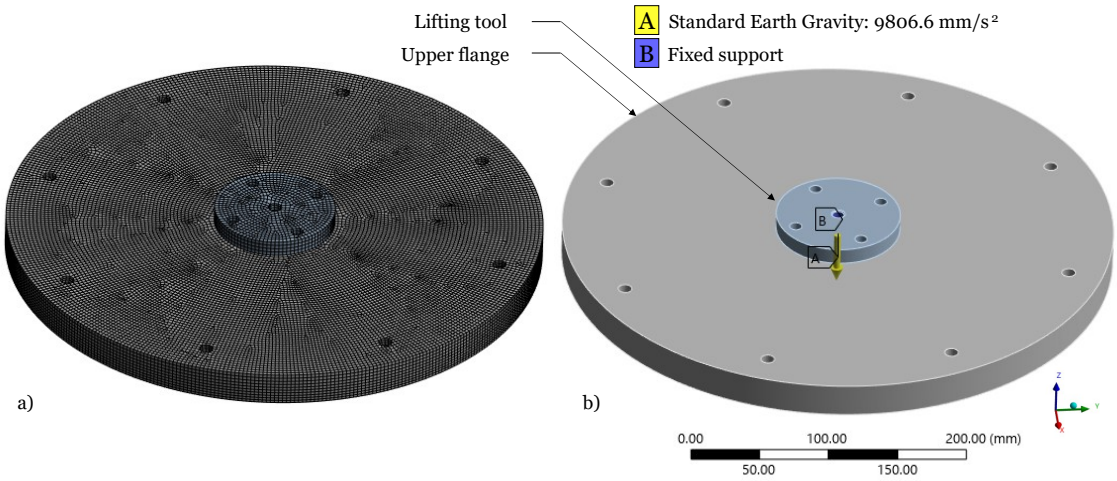

Figure 31: Upper flange calculation. a) The mesh has been set with a hex dominant method with a 3 mm maximum size. b) The boundary conditions are set as follows: the surface of the central hole, where the lifting eye will be screwed, are bounded: $U_x = U_y = U_z = 0$. The assembly is under earth gravity along the -Z axis. The contact between the lifting tool and the upper flange is assumed to be perfect. Since the sum of the section of the four holes fixing the lifting tool to upper flange (4x M8) is greater than the unique M10 section of the lifting eye, this latest is the limiting factor. The material is set with the standard structural steel from ANSYS, which is well corresponding to either the SIS2333 of the material of the atomizer, and the C-15 of the lifting eye.

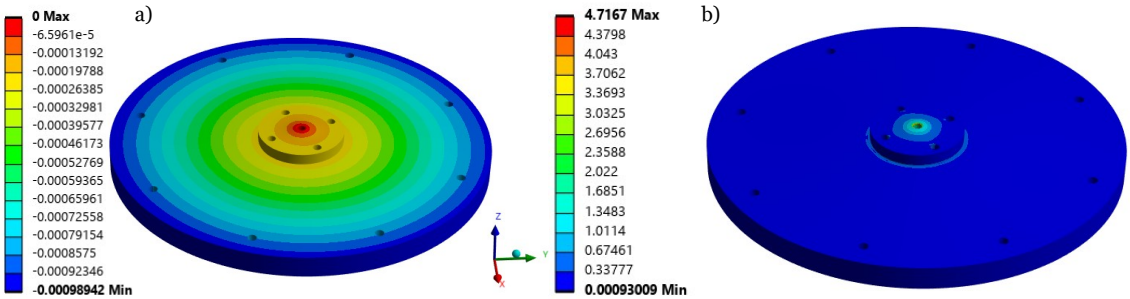

Figure 32: Upper flange results. a) The vertical (Z-axis) displacement (mm). b) The Von-Mises equivalent stress (MPa).

The maximum vertical displacement is around almost zero as expected since the thickness of the upper flange is 25 mm with a ∅ 400 mm, which is a first information showing the stiffness of the component. The maximum Von-Mises stress is around 5 MPa, which is very low compared to 133 MPa, which is 2/3 of a 200 MPa, of usual security factor applied to a very low yield strength for steels, to be conservative. Then, the design of the lifting tool and the upper flange seems to be very securely dimensioned for the expected load, even is another security factor of 5 is applied as it is often the case for lifting tools.

The lifting tool will be fixed with four M8 screws, which are more than enough to support the load of 25 kg. The four M8 threaded holes dedicated for the lifting tool (see Figure 33 b)) are not

through holes to not create a leak path for the vacuum and inert gas, but they are deep enough to ensure a good fixation of the lifting tool on the upper flange. Also they are located on a circle big enough to let the o-ring sealing surface free for the stopper rod system.

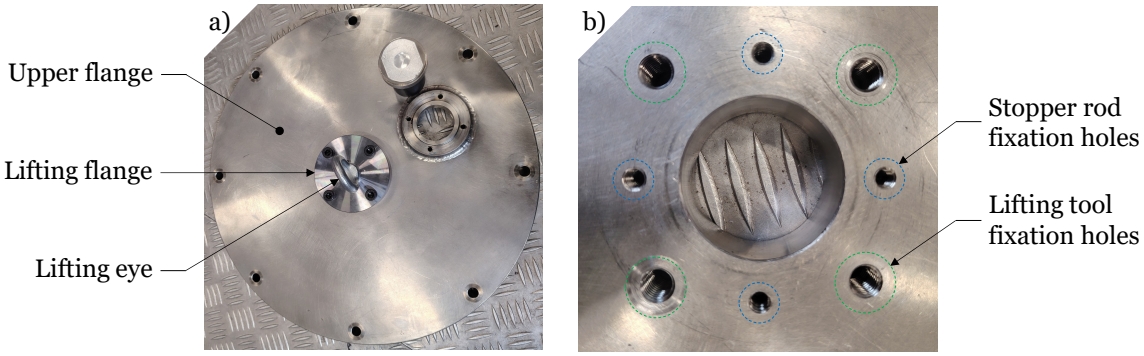

Figure 33: Upper flange with its lifting tool. a) The upper flange. b) The fixation holes in the upper flange dedicated for the stopper rod system and the lifting tool

7.3 Optical thermal sensor electric box

2026/03/25

The optical thermal sensor is composed of four main components shown in Figure 15. For a more convenient use, the power supply and the USB data converter have been put together in a box, as shown in Figure 34. Some worn-out connection have been replaced (see Figure 34 b) and c)).

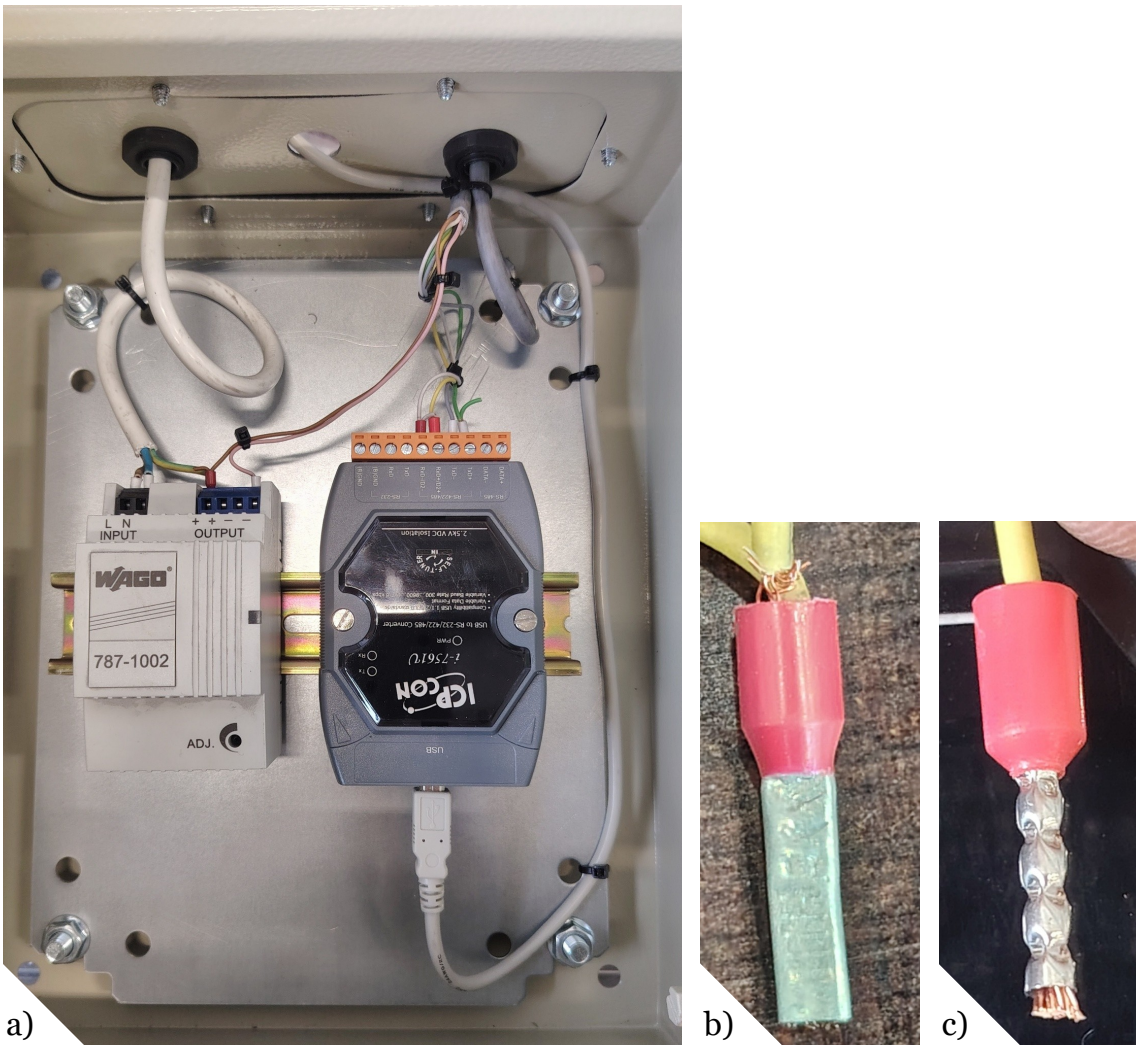

Figure 34: Electric box for the optical thermal sensor. a) The box with the power supply and the USB data converter. b) One of the worn -out connection c) A new connection.

7.4 Oxygen sensor calibration

The system contains two oxygen sensors, one in the heating chamber and one in the atomization chamber. These sensors are used to measure the oxygen level in the chambers. Combined with the pressure sensors, they allow to be sure that the atomizer atmosphere is only inert gaz with a pressure which is controlled just above the 1 bar. All the O-rings of the system ensure a quite small leak rate, which is imposed from inside to outside thanks to the overpressure. Then, the level of oxygen measured inside the atomizer is expected to be only the amount of oxygen theoretically present in the inter gaz. However, these oxygen sensors have been calibrated in 2020 and have to be re-calibrated.

These sensors are called "OXY1" and "OXY2" iun the atomizer, for respectively the heating chamber and the atomization chamber.

PZA-MC25-N/P from PEWATRON.

7.5 New melt nozzle concept

Carbotech brand proposes some sintered alumina tubes "AluSIK-60 ZA".

Order No.	Diameter (mm)
R 200	1 x 0.5
R 201	1.5 x 0.8
R 202	2 x 1
R 203	3 x 1.5
R 204	4 x 2
R 006	5 x 3
R 007	6 x 4

Table 4: Maximum temp = 1600° C, Bulk density = 2.7 g/cm^3, thermal expansion coefficient = 5.5 $\times 10^{-6} K^{-1}$, thermal conductivity = 5 W/mK

Carbotech brand proposes some sintered alumina tubes "AluSIK-99 ZA".

Order No.	Diameter (mm)
R 200	1 x 0.5
R 201	1.5 x 0.8
R 202	2 x 1
R 203	3 x 1.5
R 204	4 x 2
R 205	5 x 3
R 206	6 x 3
R 207	7 x 4

Table 5: Maximum temp = 1750° C, Bulk density = 3.8 g/cm^3, thermal expansion coefficient = 5 $\times 10^{-6} K^{-1}$, thermal conductivity = 7 W/mK

TO-DO LIST

≻ Departemental work hours have to be registred in the DTU system, how?, when? 5
≻ ———— 5
≻ ———— 5
≻ Why: gas jets direction are generally tangential to the liquid metal stream 8
≻ PhD Daniel, Figure 2.9: Effect of temperature gradient G and growth rate R on the morphology and size of solidification microstructure. Reproduced from [83]. 18
≻ Increase the temperature of the gas? 29
≻ Create a circular flow in the chamber so that the droplet takes all the same direction = less satellites? 29
≻ gas speed / distance out of nozzle Niels PhD p16-17 31
≻ Go to Daniel PhD many info about that 33
≻ ———— 34
≻ ———— 34
≻ ———— 40
≻ How to calibrate these sensors? 41
≻ ———— 42
≻ ———— 42
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≻ ———— 55

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